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Showerhead and bubble generating unit

Abstract

A shower head to jet an air bubble-liquid mixture by mixing air bubbles into the liquid includes a shower nozzle, a flow-adjustment piece, and air introduction passages. The shower nozzle includes an air bubble mixing space. The flow-adjustment piece is arranged in the air bubble mixing space. The air introduction passages cause air to flow into the air bubble mixing space therethrough.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application to U.S. application Ser. No. 16/981,443, filed on Oct. 14, 2020 and entitled SHOWERHEAD AND BUBBLE GENERATING UNIT, the entire contents of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present invention relates to a shower head configured to generate an air bubble-liquid mixture by mixing the air (air bubbles) into a liquid, or form a liquid into a mist of liquid droplets in which air bubbles are mixed, and jet the air bubble-liquid mixture or the mist of liquid droplets.

BACKGROUND ART

(2) As a technology of mixing the air into a liquid, Patent Literature 1 discloses a shower apparatus. In the shower apparatus, the liquid is jetted through a plurality of nozzle portions into a reduced tapered portion. When the liquid is jetted through the nozzle portions, the air is introduced through air inlets into the reduced tapered portion.

(3) In the shower apparatus of Patent Literature 1, as a result of collision of the liquid and the air with the reduced tapered portion, air bubbles are mixed into the liquid.

CITATION LIST

Patent Literature

(4) [PTL 1] Japanese Unexamined Patent Application Publication No. 2002-102100

SUMMARY OF INVENTION

Technical Problem

(5) However, in Patent Literature 1, as a result of collision of the liquid and the air with the reduced tapered portion, the air bubbles are mixed into the liquid. Thus, there is the possibility that a sufficient volume of the air bubbles cannot be mixed into the liquid.

(6) The present invention provides a shower head capable of mixing a sufficient volume of air bubbles into a liquid.

(7) The present invention provides a shower head configured to form a liquid into a mist of liquid droplets in which air bubbles are mixed.

Solution to Problem

(8) According to a first aspect of the present invention, there is provided a shower head including a shower main body including an inflow passage into which a liquid is caused to flow, and an outflow passage through which the liquid having flowed into the inflow passage is caused to flow out, the inflow passage being opened to one end of the shower main body, the outflow passage being opened to the other end of the shower main body; a shower nozzle mounted to the other end of the shower main body, the shower nozzle including a shower nozzle plate; a shower cylindrical portion, which has one cylinder end closed by the shower nozzle plate, is protruded to the outflow passage side, and defines an air bubble mixing space into which the liquid flowed out through the outflow passage is caused to flow from the other cylinder end of the shower cylindrical portion; and a plurality of air bubble-liquid mixture jetting holes formed in the shower nozzle plate so as to be opened in the air bubble mixing space and configured to cause an air bubble-liquid mixture to jet out of the air bubble mixing space therethrough; and air bubble-liquid mixture generating means configured to generate the air bubble-liquid mixture by mixing the air into the liquid, the air bubble-liquid mixture generating means including a flow-adjustment piece arranged in the air bubble mixing space in the shower cylindrical portion; and a plurality of air introduction passages formed in the shower nozzle, and configured to cause the air to flow into the air bubble mixing space therethrough, the flow-adjustment piece including a flow-adjustment nozzle disk arranged in the air bubble mixing space at a distance from the shower nozzle plate, and fixed to the shower cylindrical portion so as to close the other cylinder end of the shower cylindrical portion; a plurality of flow-adjustment-piece plates formed on the flow-adjustment nozzle disk, and arranged in the air bubble mixing space between the shower nozzle plate and the flow-adjustment nozzle disk; and a

plurality of liquid throttle holes formed in a portion of the flow-adjustment nozzle disk between the flow-adjustment-piece plates, and configured to cause the liquid flowed out through the outflow passage to jet into the air bubble mixing space therethrough, wherein the liquid throttle holes are formed to pass through the flow-adjustment nozzle disk so that a hole center line of each of the liquid throttle holes is arranged in parallel to a cylinder center line of the shower cylindrical portion, wherein the flow-adjustment-piece plates are protruded from the flow-adjustment nozzle disk toward the shower nozzle, and are arranged with a mixing gap separating from the shower nozzle plate, wherein the flow-adjustment-piece plates are arranged to extend from a plate center line of the flow-adjustment nozzle disk toward the shower cylindrical portion, wherein each of the flow-adjustment-piece plates causes the liquid jetted through the liquid throttle holes to flow turbulently and flow into the mixing gap on a protruding end side protruding toward the shower nozzle, wherein the air introduction passages are opened in the shower nozzle, and wherein the air introduction passages are formed to pass through the shower cylindrical portion between the protruding end of each of the flow-adjustment-piece plates and the flow-adjustment nozzle disk in a direction orthogonal to the cylinder center line of the shower cylindrical portion and are opened into the air bubble mixing space.

(9) According to a second aspect of the present invention, in the shower head according to the first aspect described above, the flow-adjustment-piece plates are arranged at equal intervals in the circumferential direction of the flow-adjustment nozzle disk.

(10) According to a third aspect of the present invention, in the shower head according to the first aspect described above, the flow-adjustment piece includes four flow-adjustment-piece plates, and the four flow-adjustment-piece plates are arranged at equal intervals in the circumferential direction of the flow-adjustment nozzle disk.

(11) According to a fourth of the present invention, in the shower head according to any one of the first to third aspects described above, the flow-adjustment-piece plates are each formed into a rectangular shape, and the flow-adjustment-piece plates each include flow-adjustment flat surfaces each formed into a rectangular shape so as to be parallel to each other with an interval equal to a thickness of each of the flow-adjustment-piece plates in the circumferential direction of the flow-adjustment nozzle disk, and a flow inclined surface formed to incline and extend from the protruding end of each of the flow-adjustment-piece plates toward one of the flow-adjustment flat surfaces and the flow-adjustment nozzle disk.

(12) According to a fifth of the present invention, in the shower head according to any one of the first to fourth aspects described above, the plurality of liquid throttle holes are arranged at equal intervals on each of a plurality of circles having different radii with a plate center line of the flow-adjustment nozzle disk being a center.

(13) According to a sixth aspect of the present invention, in the shower head according to any one of the first to fifth aspects described above, the air introduction passages are arranged at equal intervals in the circumferential direction of the shower cylindrical portion.

(14) According to a seventh aspect of the present invention, in the shower head according to any one of the first to sixth aspects described above, the air introduction passages are adjacent to the flow-adjustment nozzle disk, and are opened into the air bubble mixing space.

(15) In a seventh aspect described above, the following configuration may also be adopted. Specifically, the air introduction passages are arranged at equal intervals in the circumferential direction of the shower cylindrical portion. Further, the air introduction passages each have a flow passage width larger than a plate width of each of the flow-adjustment-piece plates in the circumferential direction of the shower cylindrical portion, and are opened into the air bubble mixing space.

(16) According to an eighth aspect of the present invention, in the shower head according to any one of the first to the seventh aspects described above, the shower head further includes flow passage switching means arranged between the air bubble-liquid mixture generating means and the

outflow passage and in the outflow passage of the shower main body; and mist generating means arranged on the shower nozzle plate on an outer side of the air bubble-liquid mixture jetting holes, and configured to form the liquid, which is caused to flow into the mist generating means through the flow passage switching means, into a mist of liquid droplets, the mist generating means including a plurality of mist throttle holes, which are formed to pass through the shower nozzle plate on the outer side of the air bubble-liquid mixture jetting holes, and are opened between the shower nozzle plate and the flow passage switching means; and a plurality of mist guides, which are each formed into a conical spiral shape, and each include a plurality of spiral surfaces each having the same spiral shape, the mist throttle holes are each formed into a conical hole passing through the shower nozzle plate and having a diameter gradually reducing from the outflow passage side, the spiral surfaces are arranged between a cone bottom flat surface and a cone upper surface of each of the mist guides to cross a cone side surface of each of the mist guides, and are each formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface, each of the mist guides is inserted into each of the mist throttle holes from the cone upper surface with a gap between the cone side surface and a conical inner peripheral surface of each of the mist throttle holes, each of the mist guides is fitted in each of the mist throttle holes so as to define a plurality of mist flow passages each having a spiral shape between the spiral surfaces and the conical inner peripheral surface, the mist flow passages are opened into each of the mist throttle holes, and are opened between the shower nozzle and the flow passage switching means, and the flow passage switching means allows connection between the liquid throttle holes and the outflow passage, or allows connection between the mist throttle holes and the outflow passage.

(17) According to a ninth aspect of the present invention, in the shower head according to the eighth aspect described above, the mist generating means includes a plurality of mist guides, which are each formed into a conical spiral shape, and each include first and second spiral surfaces each having the same spiral shape, the first and second spiral surfaces are arranged between the cone bottom flat surface and the cone upper surface to cross the cone side surface of each of the mist guides, the first and second spiral surfaces are arranged so as to be point symmetrical with respect to a cone center line of each of the mist guides, the first and second spiral surfaces are each formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface, each of the mist guides is inserted into each of the mist throttle holes from the cone upper surface with the gap between the cone side surface and the conical inner peripheral surface of each of the mist throttle holes, each of the mist guides defines first and second mist flow passages each having a spiral shape between the first and second spiral surfaces and the conical inner peripheral surface, and the first and second mist flow passages are opened in each of the mist throttle holes, and are opened between the shower nozzle and the flow passage switching means.

(18) According to a tenth aspect of the present invention, in the shower head according to the eighth or ninth aspect described above, the mist throttle holes are arranged at equal intervals on a circle that has a center along the cylinder center line of the shower cylindrical portion and is located on the outer side of the air bubble-liquid mixture jetting holes.

(19) According to an eleventh of the present invention, in the shower head according to the tenth aspect described above, the mist generating means includes a guide ring having a radius equal to a radius of the circle on which the mist throttle holes are arranged, the mist guides are arranged at equal intervals in the circumferential direction of the guide ring, each of the mist guides is fixed integrally with the guide ring so that the cone bottom flat surface is abutted on the guide ring, the guide ring is externally fitted to the shower cylindrical portion from the other cylinder end, and is arranged on the outer side of the air bubble-liquid mixture jetting holes, and, along with the insertion of the mist guides into the mist throttle holes, the guide ring is brought into abutment against the shower nozzle plate from the outflow passage side.

(20) According to a twelfth aspect of the present invention, there is provided a shower head,

including a shower main body including an inflow passage into which a liquid is caused to flow, and an outflow passage through which the liquid having flowed into the inflow passage is caused to flow out, the inflow passage being opened to one end of the shower main body, the outflow passage being opened to the other end of the shower main body; a shower nozzle mounted to the other end of the shower main body; and mist generating means arranged on the shower nozzle, and configured to form the liquid having flowed out through the outflow passage into a mist of liquid droplets, the mist generating means including a plurality of mist throttle holes, which are formed to pass through the shower nozzle, and communicate with the outflow passage; and a plurality of mist guides, which are each formed into a conical spiral shape, and each include a plurality of spiral surfaces having the same spiral shape, wherein the mist throttle holes are each formed into a conical hole passing through the shower nozzle and having a diameter gradually reducing from the outflow passage side, wherein the spiral surfaces are arranged between a cone bottom flat surface and a cone upper surface of each of the mist guides to cross a cone side surface of each of the mist guides, and are each formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface, wherein each of the mist guides is inserted into each of the mist throttle holes from the cone upper surface with a gap between the cone side surface and a conical inner peripheral surface of each of the mist throttle holes, wherein each of the mist guides is fitted in each of the mist throttle holes so as to define a plurality of mist flow passages each having a spiral shape between the spiral surfaces and the conical inner peripheral surface, and wherein the mist flow passages are opened into each of the mist throttle holes, and communicate with the outflow passage.

(21) According to a thirteenth aspect of the present invention, in the shower head according to the twelfth aspect described above, the mist generating means includes a plurality of mist guides, which are each formed into a conical spiral shape, and each include first and second spiral surfaces each having the same spiral shape, the first and second spiral surfaces are arranged between the cone bottom flat surface and the cone upper surface to cross the cone side surface of each of the mist guides, the first and second spiral surfaces are arranged so as to be point symmetrical with respect to a cone center line of each of the mist guides, the first and second spiral surfaces are each formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface, each of the mist guides is inserted into each of the mist throttle holes from the cone upper surface with the gap between the cone side surface and the conical inner peripheral surface of each of the mist throttle holes, each of the mist guides defines first and second mist flow passages each having a spiral shape between the first and second spiral surfaces and the conical inner peripheral surface, and the first and second mist flow passages are opened into each of the mist throttle holes, and communicate with the outflow passage.

Advantageous Effects of Invention

(22) According to the first aspect of the present invention, the liquid is caused to flow into the inflow passage from the one end of the shower main body, and the liquid is caused to flow through the inflow passage into the outflow passage. The liquid is caused to flow out through the outflow passage into the liquid throttle holes of the flow-adjustment piece. Through the liquid throttle holes, the liquid having flowed out through the outflow passage is jetted into the air bubble mixing space. Through the liquid throttle holes, the liquid is jetted into the air bubble mixing space toward the shower nozzle plate. In the air bubble mixing space (or in the shower cylindrical portion), the liquid is jetted between the shower nozzle and the flow-adjustment nozzle disk while flowing (being adjusted in flow) in parallel to the cylinder center line of the shower cylindrical portion.

(23) When the liquid is jetted into the air bubble mixing space, due to the flow of the liquid, the air is introduced through the air introduction passages into the air bubble mixing space. The air is caused to flow (jet) into the air bubble mixing space between the protruding ends of the flow-adjustment-piece plates and the flow-adjustment nozzle disk. The air is caused to flow (jet) between the flow-adjustment-piece plates in the air bubble mixing space.

(24) The liquid jetted through the liquid throttle holes, and the air caused to flow (jet) out through the air introduction passages are mixed in the air bubble mixing space. In the air bubble mixing space, the liquid and the air are caused to flow turbulently on the protruding end side of each of the flow-adjustment-piece plates, and flow into the mixing gap between the flow-adjustment-piece plates and the shower nozzle plate.

(25) Thus, in the mixing gap within the air bubble mixing space, due to the turbulent flow, the air mixed into the liquid is broken (divided) into micrometer-sized air bubbles (microbubbles) and nanometer-sized air bubbles (ultrafine bubbles).

(26) The micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) mix with and dissolve in the liquid.

(27) The air bubble-liquid mixture, in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed, is jetted from the air bubble-liquid mixture jetting holes to the outside.

(28) As described above, according to the first aspect of the present invention, the liquid throttle holes, the flow-adjustment-piece plates, and the air introduction passages of the flow-adjustment piece allow a sufficient volume of the micrometer-sized and nanometer-sized air bubbles (microbubbles and ultrafine bubbles) to be mixed and dissolved into the liquid.

(29) In international standards “ISO20480-1” by International Organization for Standardization (ISO), an air bubble of from equal or greater than one micrometer to a hundred micrometer (μm) is defined as a “microbubble”, and an air bubble of less than one micrometer is defined as an “ultrafine bubble” (the same applies in the following description).

(30) According to the second aspect of the present invention, the liquid is jetted from the liquid throttle holes toward between the flow-adjustment-piece plates.

(31) According to the third aspect of the present invention, the liquid is jetted equally from the liquid throttle holes toward between the four flow-adjustment-piece plates. The four flow-adjustment-piece plates allow a sufficient volume of the micrometer-sized and nanometer-sized air bubbles (microbubbles and ultrafine bubbles) to be mixed and dissolved into the liquid.

(32) According to the fourth aspect of the present invention, the flow inclined surfaces of the flow-adjustment-piece plates lead the liquid (adjusted in flow) jetted from the liquid throttle holes to the protruding ends of the flow-adjustment-piece plates, which allows the liquid and the air to flow turbulently into the mixing gap.

(33) According to the fifth of the present invention, the liquid is jetted equally from each of the liquid throttle holes throughout the air bubble mixing space.

(34) According to the sixth aspect of the present invention, the air is equally flowed (jetted) out between the flow-adjustment-piece plates through the air introduction passages.

(35) According to the seventh aspect of the present invention, the air is flowed out into the air bubble mixing space from each of the air introduction passages adjacent to the flow-adjustment nozzle disk, which allows the air to be mixed into the liquid at the same time as the liquid is jetted from the liquid throttle holes.

(36) According to the eighth aspect of the present invention, the flow passage switching means allows connection (communication) between the liquid throttle holes and the outflow passage, or allows connection (communication) between the mist throttle holes and the outflow passage.

(37) The mist throttle holes and the outflow passages are connected to flow the liquid from the one end of the shower main body into the inflow passage and to flow the liquid from the inflow passage into the outflow passage. The liquid is caused to flow through the outflow passage into the mist throttle holes. In the mist throttle holes, the liquid is caused to flow through the mist flow passages having a spiral shape into the mist throttle holes. Further, the mist of liquid droplets is jetted from each of the mist throttle holes to the outside through the mist throttle holes.

(38) The liquid is increased in pressure by flowing through the mist flow passages having a spiral shape, and is jetted into the mist throttle holes through the mist flow passages. Thus, the liquid

jetted through the mist flow passages into the mist throttle holes flows turbulently at high pressure. Further, when the mist of liquid droplets is jetted from the mist throttle holes, an outlet side of each of the mist throttle holes (a side from which the mist of liquid droplets is jetted) is brought into a negative pressure state.

(39) With the outlet side of each of the mist throttle holes brought into the negative pressure state, when the liquid, which is jetted into the mist throttle holes through the mist flow passages and flows turbulently at high pressure, passes through the outlet portion of each of the mist throttle holes, the air bubbles are separated out due to reduced pressure, and the air that is taken in at the time of jetting is broken up (divided) by the turbulent flow. Thus, the liquid is formed into the mist of liquid droplets in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved.

(40) The mist of liquid droplets in which the air bubbles are mixed is jetted from the mist throttle holes to the outside.

(41) According to the eighth aspect described above, the mist guides and the mist throttle holes allow the mist of liquid droplets in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved to be jetted to the outside.

(42) According to the ninth aspect of the present invention, the plurality of minimum mist flow passages (spiral surfaces) allow the liquid to be formed into a sufficient mist of liquid droplets. With point symmetrical arrangement of the first and second spiral surfaces, the first and second mist flow passages are arranged so as to be opposed to (face to face) each other at the cone upper surface.

(43) With this arrangement, the high-pressure liquid jetted into each of the mist throttle holes through the first and second mist flow passages is caused to collide with the cone upper surface and thereby is formed into the mist of liquid droplets in which a sufficient volume of the micrometer-sized air bubbles (microbubbles) and a sufficient volume of the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved.

(44) According to the tenth aspect of the present invention, the liquid having flowed out through the outflow passage is distributed equally in the peripheral direction of the shower cylindrical portion, and is flowed into the mist throttle holes (or into the mist flow passages).

(45) According to the eleventh aspect of the present invention, the mist guides are fixed to the guide ring, which prevents the mist guides from getting into the mist throttle holes due to the flow of the liquid even when the liquid is flowed through the outflow passage into the mist throttle holes.

(46) According to the twelfth aspect of the present invention, the liquid is flowed from the one end of the shower main body into the inflow passage, and the liquid is caused to flow through the inflow passage into the outflow passage. The liquid is caused to flow through the outflow passage into the mist throttle holes. In the mist throttle holes, the liquid is caused to flow through the mist flow passages each having a spiral shape into the mist throttle holes. Further, the mist of liquid droplets is jetted from the mist throttle holes to the outside.

(47) The liquid is increased in pressure by flowing through the mist flow passages each having a spiral shape, and is jetted into the mist throttle holes through the mist flow passages. Thus, the liquid jetted through the mist flow passages into the mist throttle holes flows turbulently at high pressure. Further, when the mist of liquid droplets is jetted from the mist throttle holes, an outlet side of each of the mist throttle holes (a side from which the mist of liquid droplets is jetted) is brought into a negative pressure state.

(48) With the outlet side of each of the mist throttle holes brought into the negative pressure state, when the liquid, which is jetted into the mist throttle holes through the mist flow passages and flows turbulently at high pressure, passes through the outlet portion of each of the mist throttle holes, the air bubbles are separated out due to reduced pressure, and the air that is taken in at the time of jetting is broken (divided) by the turbulent flow. Thus, the liquid is formed into the mist of

liquid droplets in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved.

(49) The mist of liquid droplets in which the air bubbles are mixed is jetted from the mist throttle holes to the outside.

(50) According to the twelfth aspect described above, the mist guides and the mist throttle holes allow the mist of liquid droplets in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved can be jetted to the outside.

(51) According to the thirteenth aspect of the present invention, the plurality of minimum mist flow passages (spiral surfaces) allow the liquid to be formed into a sufficient mist of liquid droplets. With point symmetrical arrangement of the first and second spiral surfaces, the first and second mist flow passages are arranged so as to be opposed to (face to face) each other at the cone upper surface.

(52) With this arrangement, the high-pressure liquid jetted into each of the mist throttle holes through the first and second mist flow passages is caused to collide with the cone upper surface, and thereby is formed into the mist of liquid droplets in which a sufficient volume of the micrometer-sized air bubbles (microbubbles) and a sufficient volume of the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view showing a shower head (at a shower position P1).
- (2) FIG. 2 is a sectional view taken along the arrows A-A in FIG. 1 (at the shower position P1).
- (3) FIG. 3 is a view seen from a direction indicated by the arrows B-B in FIG. 2 (at the shower position).
- (4) FIG. 4 is an exploded perspective view of the shower head illustrating a shower main body, flow passage switching means (including a switching handle, a switching base, a sealing gasket, sealing rings, a switching valve seat element, a switching valve element, a fixing screw bolt, and a coil spring), a shower nozzle, air bubble-liquid mixture generating means (a flow-adjustment piece), and mist generating means (including mist guides and a guide ring).
- (5) FIG. 5 is a front view showing the shower main body.
- (6) FIG. 6 is a side view showing the shower main body.
- (7) FIG. 7 is a top view showing the shower main body.
- (8) FIG. 8 is a sectional view taken along the arrows C-C in FIG. 7.
- (9) FIG. 9(a) is a top perspective view showing the switching handle of the flow passage switching means. FIG. 9(b) is a bottom perspective view of the switching handle of the flow passage switching means.
- (10) FIG. 10 is a top view showing the switching handle of the flow passage switching means.
- (11) FIG. 11(a) is a side view showing the switching handle of the flow passage switching means.
- (12) FIG. 11(b) is a sectional view taken along the arrows D-D of FIG. 10.
- (13) FIG. 12 is a bottom view showing the switching handle of the flow passage switching means.
- (14) FIG. 13(a) is a top perspective view showing the switching base of the flow passage switching means. FIG. 13(b) is a bottom perspective view showing the switching base of the flow passage switching means.
- (15) FIG. 14(a) is a top view showing the switching base of the flow passage switching means.
- (16) FIG. 14(b) is a bottom view showing the switching base of the flow passage switching means.
- (17) FIG. 15(a) is a side view showing the switching base of the flow passage switching means.
- (18) FIG. 15(b) is a sectional view taken along the arrows E-E of FIG. 14(a).

(19) FIG. **16(a)** is a top perspective view showing the switching valve seat element of the flow passage switching means. FIG. **16(b)** is a bottom perspective view showing the switching valve seat element of the flow passage switching means.

(20) FIG. **17(a)** is a top view showing the switching valve seat element of the flow passage switching means. FIG. **17(b)** is a bottom view showing the switching valve seat element of the flow passage switching means.

(21) FIG. **18(a)** is a side view showing the switching valve seat element of the flow passage switching means. FIG. **18(b)** is a sectional view taken along the arrows F-F of FIG. **17(a)**.

(22) FIG. **19(a)** is a top perspective view showing the switching valve element of the flow passage switching means. FIG. **19(b)** is a bottom perspective view showing the switching valve element of the flow passage switching means.

(23) FIG. **20** is a top view showing the switching valve element of the flow passage switching means.

(24) FIG. **21** are views showing the switching valve element of the flow passage switching means, in which FIG. **21(a)** is a bottom view illustrating a relationship between respective cylindrical valve elements, and FIG. **21(b)** is a bottom view illustrating a relationship between first regulating protruding portions and second handle regulating protruding portions.

(25) FIG. **22** are views showing the switching valve element of the flow passage switching means, in which FIG. **22(a)** is a side view seen from the first handle regulating protruding portions, and FIG. **22(b)** is a side view seen from the second handle regulating protruding portions.

(26) FIG. **23** is a sectional view taken along the arrows G-G of FIG. **20**.

(27) FIG. **24** are views showing the switching valve element of the flow passage switching means, in which FIG. **24(a)** is a sectional view taken along the arrows H-H of FIG. **20**, and FIG. **24(b)** is a sectional view taken along the arrows I-I of FIG. **20**.

(28) FIG. **25** is a sectional view taken along the arrows J-J of FIG. **22(b)**.

(29) FIG. **26** is a top view showing a handle unit (including the switching handle and the switching base) of the flow passage switching means.

(30) FIG. **27** is a bottom view showing the handle unit (including the switching handle and the switching base) of the flow passage switching means.

(31) FIG. **28** is a side view showing the handle unit (including the switching handle and the switching base) of the flow passage switching means.

(32) FIG. **29** is a sectional view taken along the arrows K-K of FIG. **26**.

(33) FIG. **30** is an enlarged sectional view illustrating a state in which the handle unit (including the switching handle and the switching base) of the flow passage switching means is arranged in the shower main body.

(34) FIG. **31** is a view seen from a direction indicated by the arrows L-L of FIG. **30**.

(35) FIG. **32** is a sectional view taken along the arrows M-M of FIG. **30**.

(36) FIG. **33** is an enlarged sectional view illustrating a state in which the fixing screw bolt and the coil spring of the flow passage switching means are arranged in the shower main body.

(37) FIG. **34** is a view seen from a direction indicated by the arrows N-N of FIG. **33**.

(38) FIG. **35** is an enlarged sectional view illustrating a state in which the switching valve seat element of the flow passage switching means is arranged in the switching base (or in the shower main body).

(39) FIG. **36** is a view seen from a direction indicated by the arrows O-O of FIG. **35**.

(40) FIG. **37** is a sectional view taken along the arrows P-P of FIG. **35**.

(41) FIG. **38** is an enlarged sectional view illustrating a state in which the switching valve element of the flow passage switching means is arranged in the switching handle (or in the shower main body).

(42) FIG. **39** is a view seen from a direction indicated by the arrows Q-Q of FIG. **38**.

(43) FIG. **40** is a sectional view taken along the arrows R-R of FIG. **38**.

(44) FIG. 41 is a sectional view taken along the arrows S-S of FIG. 38.

(45) FIG. 42(a) is a top perspective view showing the shower nozzle.

(46) FIG. 42(b) is a bottom perspective view showing the shower nozzle.

(47) FIG. 43(a) is a top view showing the shower nozzle.

(48) FIG. 43(b) is a partial enlarged view of FIG. 43(a).

(49) FIG. 44(a) is a side view showing the shower nozzle.

(50) FIG. 44(b) is a sectional view taken along the arrows T-T of FIG. 43(a).

(51) FIG. 45 is a bottom view showing the shower nozzle.

(52) FIG. 46 are views for illustrating the flow-adjustment piece of the air bubble-liquid mixture generating means, in which

(53) FIG. 46(a) is a top perspective view showing the flow-adjustment piece of the air bubble-liquid mixture generating means. FIG. 46(b) is a bottom perspective view showing the flow-adjustment piece of the air bubble-liquid mixture generating means.

(54) FIG. 47(a) is a top view showing the flow-adjustment piece of the air bubble-liquid mixture generating means. FIG. 47(b) is a partial enlarged view of FIG. 46(a).

(55) FIG. 48 are views showing the flow-adjustment piece of the air bubble-liquid mixture generating means, in which FIG. 48(a) is a top perspective view showing flow-adjustment-piece plates and flow inclined surfaces, FIG. 48(b) is a side view, and FIG. 48(c) is a partial enlarged view of FIG. 48(b).

(56) FIG. 49(a) is a bottom view showing the flow-adjustment piece of the air bubble-liquid mixture generating means. FIG. 49(b) is a sectional view taken along the arrows U-U of FIG. 47(a).

(57) FIG. 50 are views illustrating a state in which the flow-adjustment piece is incorporated in the shower nozzle, in which FIG. 50(a) is a top view, and FIG. 50(b) is a bottom view.

(58) FIG. 51 are sectional views taken along the arrows V-V of FIG. 50(a), in which FIG. 51(a) is a view illustrating a relationship between the flow-adjustment piece and a shower cylindrical portion, and FIG. 51(b) is a view illustrating a relationship between the flow-adjustment-piece plates and a shower nozzle plate.

(59) FIG. 52(a) is a top perspective view showing a mist ring body (including the guide ring and the mist guides) of the mist generating means. FIG. 52(b) is a partial enlarged view of FIG. 52(a).

(60) FIG. 53 is a bottom perspective view showing the mist ring body (including the guide ring and the mist guides) of the mist generating means.

(61) FIG. 54(a) is a top view showing the mist ring body (including the guide ring and the mist guides) of the mist generating means. FIG. 54(b) is a side view showing the mist ring body (including the guide ring and the mist guides) of the mist generating means.

(62) FIG. 55(a) is a bottom view showing the mist ring body (including the guide ring and the mist guides) of the mist generating means. FIG. 55(b) is a sectional view taken along the arrows W-W of FIG. 54(a).

(63) FIG. 56 are views illustrating a state in which the mist ring body (including the guide ring and the mist guides) is incorporated in the shower nozzle, in which FIG. 56(a) is a top view, and FIG. 56(b) is a bottom view.

(64) FIG. 57 are views illustrating a state in which the mist ring body (including the guide ring and the mist guides) is incorporated in the shower nozzle, in which FIG. 57(a) is a sectional view taken along the arrows X-X of FIG. 56(a), and FIG. 57(b) is a partial enlarged view of FIG. 57(a).

(65) FIG. 58 is a partial enlarged view of FIG. 2 (at the shower position P1).

(66) FIG. 59 is a partial enlarged view of FIG. 2 (at the shower position P1).

(67) FIG. 60 is a partial enlarged view of FIG. 59 (at the shower position P1).

(68) FIG. 61 is a perspective view showing the shower head (at a mist position P2).

(69) FIG. 62 is a partial enlarged sectional view taken along the arrows a-a of FIG. 61 (at the mist position P2).

- (70) FIG. 63 is a sectional view taken along the arrows b-b of FIG. 62 (at the mist position P2).
- (71) FIG. 64 is a sectional view taken along the arrows c-c of FIG. 62 (at the mist position P2).
- (72) FIG. 65 is a sectional view taken along the arrows d-d of FIG. 62 (at the mist position P2).
- (73) FIG. 66 is a partial enlarged view of FIG. 62 for illustrating a relationship between a mist throttle hole and the mist guide (at the mist position P2).
- (74) FIG. 67 are views showing a flow-adjustment piece in Example 1 for a “shower test”, in which FIG. 67(a) is a top view, and FIG. 67(b) is a bottom view.
- (75) FIG. 68 are views showing a flow-adjustment piece in Example 2 for the “shower test”, in which FIG. 68(a) is a top view, and FIG. 68(b) is a bottom view.
- (76) FIG. 69 are views showing a flow-adjustment piece in Example 3 for the “shower test”, in which FIG. 69(a) is a top view, and FIG. 69(b) is a bottom view.

DESCRIPTION OF EMBODIMENTS

- (77) A shower head according to the present invention is described with reference to FIG. 1 to FIG. 69.
- (78) A shower head X is configured to generate an air bubble-liquid mixture by mixing the air (air bubbles) into a liquid, or form a liquid into mist-like liquid droplets in which air bubbles are mixed, and jet the air bubble-liquid mixture or the mist-like (atomized) liquid droplets.
- (79) The liquid is water or hot water (the same applies in the following description). The air bubble-liquid mixture is air bubble-water mixture or air bubble-hot water mixture generated by mixing the air into water or hot water, or water or hot water in which microbubbles or ultrafine bubbles are mixed (the same applies in the following description).
- (80) As shown in FIG. 1 to FIG. 65, the shower head X includes a shower main body 1, flow passage switching means 2, a shower nozzle 3, air bubble-liquid mixture generating means 4, and mist generating means 5.
- (81) As shown in FIG. 1, FIG. 2, and FIG. 4 to FIG. 8, the shower main body 1 is made of a synthetic resin. The shower main body 1 includes a handle portion 6 and a head portion 7, and the handle portion 6 and the head portion 7 are formed integrally with each other. The handle portion 6 is formed into a cylindrical shape, and the head portion 7 is formed into a dome shape.
- (82) As shown in FIG. 5 to FIG. 8, the head portion 7 is arranged so that a dome top 7A side thereof is located at the other end 6B of the handle portion 6. The head portion 7 is fixed to the other end 6B of the handle portion 6 so as to be inclined toward the handle portion 6 side.
- (83) As shown in FIG. 5 and FIG. 8, the head portion 7 includes a shower space 7C and a shower cylindrical portion 8.
- (84) As shown in FIG. 5 and FIG. 8, the shower space 7C is arranged concentrically with the head portion 7, and is opened to a circular end 7B of the head portion 7 (or the other end 1B of the shower main body 1). The shower space 7C is formed to extend from the circular end 7B toward the dome top 7A side in a direction of a center line of the head portion 7. The shower space 7C is closed by the dome top 7A of the head portion 7.
- (85) As shown in FIG. 5 and FIG. 8, the shower cylindrical portion 8 is arranged in the shower space 7C. The shower cylindrical portion 8 is arranged concentrically with the shower space 7C. The shower cylindrical portion 8 is fixed on the dome top 7A side of the head portion 7 in the shower space 7C, and is formed integrally with the head portion 7. The shower cylindrical portion 8 is formed to extend from the dome top 7A side toward the circular end 7B side of the head portion 7. One cylinder end 8A of the shower cylindrical portion 8 is opened in the shower space 7C (toward the other end 1B of the shower main body 1). The other cylinder end 8B of the shower cylindrical portion 8 is closed by the dome top 7A of the head portion 7.
- (86) As shown in FIG. 5 and FIG. 8, the shower main body 1 includes an inflow passage 9, an outflow passage 10, a plurality of (three) fixing protruding portions 11, a guide protruding portion 12, a base protruding portion 13, and a reference protruding portion 14.
- (87) As shown in FIG. 5 and FIG. 8, the inflow passage 9 is a flow passage being a circular hole,

and is formed in the handle portion **6**. The inflow passage **9** is opened on one end **1A** of the shower main body **1** (or one end **6A** of the handle portion). The inflow passage **9** is formed to pass through the handle portion **6** in a direction of a cylinder center line of the handle portion **6**, and is opened on the other end **6B** of the handle portion **6**.

(88) The inflow passage **9** is opened in the outflow passage **10** on the dome top **7A** side of the head portion **7**.

(89) The one end **6A** of the handle portion **6** (or the one end **1A** of the shower main body **1**) is connected to a water supply hose (not shown), and the liquid is caused to flow through the water supply hose into the inflow passage **9**.

(90) As shown in FIG. 5 and FIG. 8, the outflow passage **10** is a flow passage being a circular hole, and is formed in the shower cylindrical portion **8** of the head portion **7**. The outflow passage **10** is opened on the other end **1B** of the shower main body **1** (or on the one cylinder end **8A** of the shower cylindrical portion **8**). The outflow passage **10** is arranged concentrically with the shower cylindrical portion **8**, and is formed to extend toward the dome top **7A** side of the head portion **7**. The outflow passage **10** is closed by the dome top **7A** of the head portion **7**. The outflow passage **10** communicates with the inflow passage **9** on the dome top **7A** side of the head portion **7**. As shown in FIG. 5 and FIG. 8, on the other end **1B** side of the shower main body **1** with respect to the inflow passage **9** (or on the one cylinder end **8A** side of the shower cylindrical portion **8**), the outflow passage **10** is reduced in diameter at a hole step portion **10A** thereof, and is formed to extend toward the dome top **7A** side of the head portion **7**.

(91) This configuration allows the liquid to flow through the inflow passage **9** into the outflow passage **10**, and the liquid is caused to flow out from the other end **1B** of the shower main body **1** (or from the circular end **7B** of the head portion **7**).

(92) As shown in FIG. 5 and FIG. 8, the plurality of fixing protruding portions **11** are arranged in the outflow passage **10**. Each of the fixing protruding portions **11** is formed to protrude from an inner peripheral surface (of the shower cylindrical portion **8**) in the outflow passage **10** toward a center line A of the outflow passage **10**, and to extend toward the dome top **7A** side of the head portion **7**. Each of the fixing protruding portions **11** is formed integrally with the inner peripheral surface of the shower cylindrical portion **8**.

(93) One of the fixing protruding portions **11** is arranged at a highest point **7a** of the head portion **7**. The other two fixing protruding portions **11** are arranged with angular intervals of 90 degrees on both side positions of the highest point **7a** in the peripheral direction (circumferential direction) of the outflow passage **10**.

(94) As shown in FIG. 5 to FIG. 8, the guide protruding portion **12** is formed into a cylindrical shape, and is formed integrally with the other end **1B** of the shower main body **1** (or the other end **7B** of the head portion **7**). The guide protruding portion **12** is arranged concentrically with the outflow passage **10**, and is formed to protrude from the other end **1B** of the shower main body **1** (or the other end **7B** of the head portion **7**).

(95) As shown in FIG. 5 and FIG. 8, the base protruding portion **13** is a column having a circular cross section, and is arranged in the outflow passage **10** of the head portion **7**. The base protruding portion **13** is arranged concentrically with the outflow passage **10**, and is supported so that one end of the base protruding portion **13** is fixed on the dome top **7A** side of the head portion **7**. The base protruding portion **13** is formed to protrude from the dome top **7A** side of the head portion **7** toward the other end **1B** of the shower main body **1** (or to the circular end **7B** of the head portion **7**) in the outflow passage **10**.

(96) The base protruding portion **13** has a screw hole **15**. As shown in FIG. 2, FIG. 5, and FIG. 8, the screw hole **15** is arranged concentrically with the outflow passage **10**, and is formed in the base protruding portion **13**. The screw hole **15** is formed to extend in a direction of the center line A of the outflow passage **10**, and is opened in the outflow passage **10**.

(97) As shown in FIG. 5 to FIG. 8, the reference protruding portion **14** is formed integrally with the

head portion **7**. The reference protruding portion **14** is arranged at the highest point **7a** of the head portion **7**. The reference protruding portion **14** is formed to protrude from a surface of the head portion **7** in a direction orthogonal to the center line A of the outflow passage **10**.

(98) As shown in FIG. **1** to FIG. **4** and FIG. **9** to FIG. **25**, the flow passage switching means **2** (flow passage switching unit) includes a switching handle **21**, a switching base **22**, a sealing gasket **23**, a sealing ring **24**, a switching valve seat element **25** (switching valve seat), a sealing ring **26**, a switching valve element **27** (switching valve), a plurality of (a pair of) sealing rings **28**, a fixing screw bolt **29**, and a coil spring **30**.

(99) As shown in FIG. **9** to FIG. **12**, the switching handle **21** is made of a synthetic resin and formed into a cylindrical shape. The switching handle **21** includes a first handle cylindrical portion **31**, a second handle cylindrical portion **32**, a handle hole **33**, a threaded portion **34**, a plurality of (a pair of) first retaining grooves **35**, a plurality of (a pair of) second retaining grooves **36**, and a handle protrusion **37**.

(100) The first handle cylindrical portion **31** (small-diameter cylindrical portion) and the second handle cylindrical portion **32** (large-diameter cylindrical portion) are arranged concentrically with each other with a cylinder center line B (a center line) of the switching handle **21** being a center, and are formed integrally with each other.

(101) The first handle cylindrical portion **31** is reduced in diameter, and extends from one cylinder end **32A** of the second handle cylindrical portion **32** in a direction of the cylinder center line B of the switching handle **21**.

(102) As shown in FIG. **9** to FIG. **12**, the second handle cylindrical portion **32** includes a shower protruding portion **38** configured to indicate a shower position P1, a mist protruding portion **39** configured to indicate a mist position P2, and a handle groove **40**.

(103) As shown in FIG. **9** to FIG. **12**, the shower protruding portion **38** and the mist protruding portion **39** are arranged with an angular interval of 90 degrees in the peripheral direction of the switching handle **21** (or the second handle cylindrical portion **32**). The shower protruding portion **38** and the mist protruding portion **39** are formed to protrude from an outer peripheral surface of the second handle cylindrical portion **32** in a direction orthogonal to the cylinder center line B of the switching handle **21**.

(104) As shown in FIG. **9(b)** and FIG. **11(b)**, the handle groove **40** is an annular groove, and is formed in the second handle cylindrical portion **32**. The handle groove **40** is arranged concentrically with the second handle cylindrical portion **32** with the cylinder center line B of the switching handle **21** being a center. The handle groove **40** is arranged on an outer side of the first handle cylindrical portion **31** in the direction orthogonal to the cylinder center line B of the switching handle **21**. The handle groove **40** is formed so as to be opened to the one cylinder end **32A** of the second handle cylindrical portion **32**.

(105) The handle groove **40** is formed to extend from the one cylinder end **32A** to the other cylinder end **32B** of the second handle cylindrical portion **32**, and has a groove depth in the direction of the cylinder center line B of the switching handle **21**.

(106) As shown in FIG. **9**, FIG. **10**, FIG. **11(b)**, and FIG. **12**, the handle hole **33** is formed into a circular hole. The handle hole **33** is arranged concentrically with the handle cylindrical portions **31** and **32** with the cylinder center line B of the switching handle **21** (including the first handle cylindrical portion **31** and the second handle cylindrical portion **32**) being a center.

(107) The handle hole **33** is formed to pass through the first handle cylindrical portion **31** and the second handle cylindrical portion **32** in the direction of the cylinder center line B of the switching handle **21**. The handle hole **33** is opened to one cylinder end **31A** of the first handle cylindrical portion **31** and the other cylinder end **32B** of the second handle cylindrical portion **32**.

(108) As shown in FIG. **9**, FIG. **10**, FIG. **11(b)**, and FIG. **12**, the handle hole **33** includes a large-diameter hole portion **33A**, a medium-diameter hole portion **33B**, and a small-diameter hole portion **33C**.

(109) The large-diameter hole portion **33A** is opened to the other cylinder end **32B** of the second handle cylindrical portion **32**. The medium-diameter hole portion **33B** is formed between the large-diameter hole portion **33A** and the small-diameter hole portion **33C**. The medium-diameter hole portion **33B** is reduced in diameter at a first hole step portion **33D** as compared to the large-diameter hole portion **33A**, and is continuous with the small-diameter hole portion **33C**.

(110) The small-diameter hole portion **33C** is reduced in diameter at a second hole step portion **33E** as compared to the medium-diameter hole portion **33B**, and is opened to the one cylinder end **31A** of the first handle cylindrical portion **31**.

(111) As shown in FIG. **9**, FIG. **10**, and FIG. **11(b)**, the threaded portion **34** is formed in the large-diameter hole portion **33A** of the handle hole **33**. The threaded portion **34** is arranged in a range from the first hole step portion **33D** to the other cylinder end **32B** side of the second handle cylindrical portion **32** in the direction of the cylinder center line B of the switching handle **21**.

(112) As shown in FIG. **9**, FIG. **10**, and FIG. **11(b)**, the first retaining grooves **35** are formed in the medium-diameter hole portion **33B** of the handle hole **33**. The first retaining grooves **35** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching handle **21** (or of the second handle cylindrical portion **32**).

(113) One of the first retaining grooves **35** is arranged at a position corresponding to the shower protruding portion **38** in the peripheral direction of the switching handle **21**.

(114) The first retaining grooves **35** are formed to extend between the first hole step portion **33D** and the second hole step portion **33E** in the direction of the cylinder center line B of the switching handle **21**. The first retaining grooves **35** each have a groove width H1 in the peripheral direction (circumferential direction) of the switching handle **21**, and are each opened in an inner peripheral surface of the medium-diameter hole portion **33B**.

(115) As shown in FIG. **9**, FIG. **10**, and FIG. **11(b)**, the second retaining grooves **36** are formed in the medium-diameter hole portion **33B** of the handle hole **33**. The second retaining grooves **36** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching handle **21** (or the second handle cylindrical portion **32**).

(116) One of the second retaining grooves **36** is arranged at a position corresponding to the mist protruding portion **39** in the peripheral direction of the switching handle **21**. The second retaining grooves **36** are each located at a center between the first retaining grooves **35** in the peripheral direction of the switching handle **21**, and are arranged with angular intervals of 90 degrees between the first retaining grooves **35**.

(117) The second retaining grooves **36** are formed to extend from the first hole step portion **33D** toward the second hole step portion **33E** side in the direction of the cylinder center line B of the switching handle **21**. The second retaining grooves **36** each have a groove width H2 in the peripheral direction of the switching handle **21**, and are each opened in the inner peripheral surface of the medium-diameter hole portion **33B**. The groove width H2 of each of the second retaining grooves **36** is smaller than the groove width H1 of each of the first retaining grooves **35** (groove width H2 < groove width H1).

(118) As shown in FIG. **9(b)**, FIG. **11**, and FIG. **12**, the handle protrusion **37** is arranged on the outer side of the first handle cylindrical portion **31** in the direction orthogonal to the cylinder center line B of the switching handle **21**. The handle protrusion **37** is arranged at a position corresponding to the shower protruding portion **38** in the peripheral direction of the switching handle **21**. The handle protrusion **37** is formed integrally on an outer peripheral surface of the first handle cylindrical portion **31**. The handle protrusion **37** is formed to protrude from the outer peripheral surface of the first handle cylindrical portion **31** to the handle groove **40** in the direction orthogonal to the cylinder center line B of the switching handle **21**.

(119) The handle protrusion **37** is formed to extend between the one cylinder end **31A** of the first handle cylindrical portion **31** and the one cylinder end **32A** of the second handle cylindrical portion **32** in the direction of the cylinder center line B of the switching handle **21**. The handle protrusion

37 includes a protrusion end surface **37A** (a flat end surface) that is flush with the one cylinder end **31A** of the first handle cylindrical portion **31**.

(120) As shown in FIG. **13** to FIG. **15**, the switching base **22** is made of a synthetic resin and formed into a cylindrical shape. The switching base **22** includes a first base cylindrical portion **45** (a large-diameter cylindrical portion), a second base cylindrical portion **46** (a small-diameter cylindrical portion), a base annular plate **47**, a base hole **48**, a fixing cylindrical portion **49**, a plurality of (a pair of) first rib portions **50**, a plurality of (a pair of) second rib portions **51**, and a plurality of (a pair of) base protrusions **59** and **60**.

(121) The first base cylindrical portion **45** and the second base cylindrical portion **46** are arranged concentrically with each other with a cylinder center line C (a center line) of the switching base **22** being a center. The first base cylindrical portion **45** and the second base cylindrical portion **46** are formed integrally with each other.

(122) As shown in FIG. **13** and FIG. **15**, the first base cylindrical portion **45** has a plurality of sealing grooves **53** and **54**.

(123) As shown in FIG. **13** and FIG. **15**, the sealing groove **53** is formed into an annular groove, and is arranged on one cylinder end **45A** side of the first base cylindrical portion **45**. The sealing groove **53** is arranged concentrically with the first base cylindrical portion **45** with the cylinder center line C (the center line) of the switching base **22** (or the first base cylindrical portion **45**) being a center. The sealing groove **53** is formed along an entire outer peripheral surface of the first base cylindrical portion **45**. The sealing groove **53** has a groove depth in a direction orthogonal to the cylinder center line C of the switching base **22**, and is opened in the outer peripheral surface of the first base cylindrical portion **45**.

(124) As shown in FIG. **13** and FIG. **15**, the sealing groove **54** is formed into an annular groove, and is arranged on the other cylinder end **45B** side of the first base cylindrical portion **45**. The sealing groove **54** is arranged between the other cylinder end **45B** of the first base cylindrical portion **45** and the sealing groove **53** in the direction of the cylinder center line C of the switching base **22**.

(125) The sealing groove **54** is arranged concentrically with the first base cylindrical portion **45** with the cylinder center line C of the switching base **22** being a center. The sealing groove **54** is formed along the entire outer peripheral surface of the first base cylindrical portion **45**. The sealing groove **54** has a groove depth in the direction orthogonal to the cylinder center line C of the switching base **22**, and is opened in the outer peripheral surface of the first base cylindrical portion **45**.

(126) As shown in FIG. **13(b)**, FIG. **14(b)**, and FIG. **15**, the second base cylindrical portion **46** is reduced in diameter at the one cylinder end **45A** of the first base cylindrical portion **45**, and is formed to protrude from the first base cylindrical portion **45** in the direction of the cylinder center line C of the switching base **22**.

(127) The second base cylindrical portion **46** has a plurality of (three) base regulating grooves **55**, **56**, and **57**.

(128) As shown in FIG. **13**, FIG. **14(b)**, and FIG. **15**, the base regulating grooves **55** to **57** are arranged with angular intervals of 90 degrees in the peripheral direction of the switching base **22**.

(129) With regard to the base regulating grooves **55** to **57**, on both sides of one base regulating groove **55** in the peripheral direction of the switching base **22**, the other two base regulating grooves **56** and **57** are arranged. Each of the base regulating grooves **56** and **57** is arranged with an angular interval of 90 degrees between the base regulating groove **55** and each of the base regulating grooves **56** and **57** in the peripheral direction of the switching base **22**.

(130) The base regulating grooves **55**, **56**, and **57** are each formed to extend between the one cylinder end **45A** of the first base cylindrical portion **45** and one cylinder end **46A** of the second base cylindrical portion **46** in the direction of the cylinder center line C of the switching base **22**, and are each opened to the one cylinder end **46A** of the second base cylindrical portion **46**.

(131) The base regulating grooves 55 to 57 each have a groove depth in the direction orthogonal to the cylinder center line C of the switching base 22, and are each opened in an outer peripheral surface of the second base cylindrical portion 46.

(132) As shown in FIG. 13 to FIG. 15, the base annular plate 47 is arranged concentrically with the first base cylindrical portion 45 with the cylinder center line C of the switching base 22 (or the first base cylindrical portion 45) being a center. The base annular plate 47 is fixed to the other cylinder end 45B of the first base cylindrical portion 45, and is formed integrally with the first base cylindrical portion 45. The base annular plate 47 is formed to protrude from the outer peripheral surface of the first base cylindrical portion 45 in the direction orthogonal to the cylinder center line C of the switching base 22.

(133) As shown in FIG. 13(a), FIG. 14, and FIG. 15(b), the base hole 48 is formed into a circular hole. The base hole 48 is formed to pass through the first base cylindrical portion 45 and the second base cylindrical portion 46 in the direction of the cylinder center line C of the switching base 22. The base hole 48 is arranged concentrically with the base cylindrical portions 45 and 46 with the cylinder center line C of the switching base 22 being a center.

(134) The base hole 48 includes a small-diameter hole portion 48A and a large-diameter hole portion 48B. The small-diameter hole portion 48A is formed to pass through the first base cylindrical portion 45, and is opened to the base annular plate 47. The large-diameter hole portion 48B is increased in diameter at a hole step portion 48C as compared to the small-diameter hole portion 48A, and is opened to the one cylinder end 46A of the second base cylindrical portion 46.

(135) As shown in FIG. 13 to FIG. 15, the fixing cylindrical portion 49 is arranged in the base cylindrical portions 45 and 46. The fixing cylindrical portion 49 is arranged concentrically with the second base cylindrical portion 46 with the cylinder center line C of the switching base 22 (including the base cylindrical portions 45 and 46) being a center.

(136) The fixing cylindrical portion 49 is arranged in the base cylindrical portions 45 and 46 with an annular space Y between the fixing cylindrical portion 49 and inner peripheral surfaces of the base cylindrical portions 45 and 46 in the direction orthogonal to the cylinder center line C of the switching base 22. The fixing cylindrical portion 49 is formed to extend from the hole step portion 48C of the base hole 48 toward the one cylinder end 46A side of the second base cylindrical portion 46 in the direction of the cylinder center line C of the switching base 22, and protrudes from the one cylinder end 46A of the second base cylindrical portion 46. The fixing cylindrical portion 49 includes a cylinder end surface 49A (a flat end surface) that is flush with the hole step portion 48C of the base hole 48.

(137) As shown in FIG. 13(b), FIG. 14, and FIG. 15(b), the fixing cylindrical portion 49 has a bolt receiving hole 58. The bolt receiving hole 58 is arranged concentrically with the fixing cylindrical portion 49 with the cylinder center line C of the switching base 22 being a center. The bolt receiving hole 58 is formed to pass through the fixing cylindrical portion 49 in the direction of the cylinder center line C of the switching base 22.

(138) As shown in FIG. 13(b), FIG. 14, and FIG. 15(b), the bolt receiving hole 58 includes a large-diameter hole portion 58A, a small-diameter hole portion 58B, and a medium-diameter hole portion 58C.

(139) With regard to the bolt receiving hole 58, the large-diameter hole portion 58A is opened to the one cylinder end surface 49A of the fixing cylindrical portion 49, and communicates with the small-diameter hole portion 48A of the base hole 48. The small-diameter hole portion 58B is arranged between the large-diameter hole portion 58A and the medium-diameter hole portion 58C. The small-diameter hole portion 58B is formed to be reduced in diameter as compared to the large-diameter hole portion 58A. The medium-diameter hole portion 58C is increased in diameter as compared to the small-diameter hole portion 58B, opened to the other cylinder end 49B of the fixing cylindrical portion 49.

(140) As shown in FIG. 13, FIG. 14, and FIG. 15(b), the first rib portions 50 are each arranged in

the large-diameter hole portion **48B** of the base hole **48** between each of the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49** (in the annular space Y). The first rib portions **50** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching base **22** (including the base cylindrical portions **45** and **46**). With regard to the first rib portions **50**, one of the first rib portions **50** is arranged at a position corresponding to the base regulating groove **55** (one of the base regulating grooves).

(141) The first rib portions **50** are each formed to extend between the hole step portion **48C** of the base hole **48** and the one cylinder end **46A** of the second base cylindrical portion **46** in the direction of the cylinder center line C of the switching base **22**. The first rib portions **50** are fixed to the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49**, and are formed integrally with the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49**. The first rib portions **50** each have a rib width hA in the peripheral direction of the switching base **22**. The first rib portions **50** each include a rib flat surface **50A** that is flush with the cylinder end surface **49A** of the fixing cylindrical portion **49** (or the hole step portion **48C**).

(142) As shown in FIG. **13**, FIG. **14** and FIG. **15(b)**, the second rib portions **51** are each arranged in the large-diameter hole portion **48B** of the base hole **48** between each of the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49** (in the annular space Y). The second rib portions **51** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching base **22** (including the base cylindrical portions **45** and **46**). The second rib portions **51** are each located at a center between the first rib portions **50** in the peripheral direction of the switching base **22**, and are arranged at positions respectively corresponding to the base regulating grooves **56** and **57** (the other two base regulating grooves). The second rib portions **51** are each formed to extend between the hole step portion **48C** of the base hole **48** and the one cylinder end **46A** of the second base cylindrical portion **46** in the direction of the cylinder center line C of the switching base **22**. The second rib portions **51** are fixed to the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49**, and are formed integrally with the base cylindrical portions **45** and **46** and the fixing cylindrical portion **49**. The second rib portions **51** each have a rib width hB in the peripheral direction of the switching base **22**. The rib width hB of each of the second rib portions **51** is larger than the rib width hA of each of the first rib portions **50** (rib width hB > rib width hA).

(143) The second rib portions **51** each include a rib flat surface **51A** that is flush with the cylinder end surface **49A** of the fixing cylindrical portion **49** (or the hole step portion **48C**).

(144) This configuration, as shown in FIG. **13(b)** and FIG. **14(b)**, in the annular space Y, allows a plurality of (four) base inflow passages Z to be defined between the first rib portions **50** and the second rib portions **51** in the peripheral direction. The base inflow passages Z are each formed to extend in the direction of the cylinder center line C of the switching base **22**, and are each opened to the large-diameter hole portion **48B** of the base hole **48** and the one cylinder end **46A** of the second base cylindrical portion **46**.

(145) As shown in FIG. **13(a)**, FIG. **14(a)**, and FIG. **15(b)**, the base protrusions **59** and **60** are fixed to the other cylinder end **45B** side of the first base cylindrical portion **45** and the base annular plate **47**, and are formed integrally with the first base cylindrical portion **45** and the base annular plate **47**.

(146) The base protrusions **59** and **60** are arranged between the base hole **48** (or the small-diameter hole portion **48A**) and the outer peripheral surface of the base annular plate **47** in the direction orthogonal to the cylinder center line C of the switching base **22**.

(147) The base protrusions **59** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching base **22**. The base protrusions **59** and **60** are arranged (in a concyclic manner) on a circle that has a center along the cylinder center line C of the switching base **22** and is located on an outer side of the base hole **48**.

(148) The base protrusions **59** and **60** are each formed to protrude from the other cylinder end **45B**

of the first base cylindrical portion **45** and the base annular plate **47** in the direction of the cylinder center line C of the switching base **22**.

(149) As shown in FIG. **14(a)**, one base protrusion **59** is arranged between the base regulating grooves **55** and **56** in the peripheral direction (circumferential direction) of the switching base **22**. The base protrusion **59** includes a first base regulating flat surface **59A** located at a base distance HA from a base longitudinal straight line LX that is orthogonal to the cylinder center line C of the switching base **22** and passes a center of the base regulating groove **55**. The first base regulating flat surface **59A** is formed in parallel to the base longitudinal straight line LX.

(150) The base protrusion **59** includes a second base regulating flat surface **59B** located at the base distance HA from a base transverse straight line LY that is orthogonal to the cylinder center line C of the switching base (or the base longitudinal straight line LX) and passes a center of each of the base regulating grooves **56** and **57**. The second base regulating flat surface **59B** is formed in parallel to the base transverse straight line LY.

(151) As shown in FIG. **14(a)**, the other one base protrusion **60** is arranged between the base regulating grooves **56** and **57** in the peripheral direction (circumferential direction) of the switching base **22**.

(152) The base protrusion **60** includes a third base regulating flat surface **60A** located at a base distance HB from the base transverse straight line LY. The third base regulating flat surface **60A** is formed in parallel to the base transverse straight line LY.

(153) The base protrusion **60** includes a fourth base regulating flat surface **60B** located at the base distance HB from the base longitudinal straight line LX. The fourth base regulating flat surface **60B** is formed in parallel to the base longitudinal straight line LX. The base distance HB is a dimension (distance) equal to the base distance HA (base distance HA=base distance HB).

(154) As shown in FIG. **4** and FIG. **15**, the sealing gasket **23** is made of an elastic material such as synthetic rubber, and is formed into an annular shape. The sealing gasket **23** is externally fitted to the first base cylindrical portion **45** of the switching base **22**, and is fitted in the sealing groove **54**. The sealing gasket **23** is arranged in the sealing groove **54** so as to protrude from the outer peripheral surface of the first base cylindrical portion **45**.

(155) As shown in FIG. **4** and FIG. **15**, the sealing ring **24** is made of an elastic material such as synthetic rubber, and is formed into an annular shape. The sealing ring **24** is externally fitted to the first base cylindrical portion **45** of the switching base **22**, and is fitted in the sealing groove **53**. The sealing ring **24** is arranged in the sealing groove **53** so as to protrude from the outer peripheral surface of the first base cylindrical portion **45**.

(156) As shown in FIG. **16** to FIG. **18**, the switching valve seat element **25** (the switching valve seat) is made of a synthetic resin and formed into a cylindrical shape. The switching valve seat element **25** includes a valve seat cylindrical portion **62**, a valve seat disk **63**, a plurality of (a pair of) valve seat holes **64** and **65**, a plurality of (a pair of) first regulating protrusions **66**, a plurality of (a pair of) second regulating protrusions **67**, and a plurality of (a pair of) spring receiving protruding portions **68**.

(157) As shown in FIG. **16**, FIG. **17(b)**, and FIG. **18**, the valve seat cylindrical portion **62** is formed into a cylindrical shape. As shown in FIG. **15(b)** and FIG. **17(a)**, an outer diameter D1 of the valve seat cylindrical portion **62** is smaller than a hole diameter d1 of the small-diameter hole portion **68A** of the base hole **48** (or the switching base **22**) (outer diameter D1<hole diameter d1).

(158) As shown in FIG. **16** to FIG. **18**, the valve seat cylindrical portion **62** has a sealing groove **69**. The sealing groove **69** is formed into an annular groove, and is arranged concentrically with the valve seat cylindrical portion **62** with a cylinder center line D (a center line) of the switching valve seat element **25** (or the valve seat cylindrical portion **62**) being a center. The sealing groove **69** is formed along an entire outer peripheral surface of the valve seat cylindrical portion **62**. The sealing groove **69** has a groove depth in a direction orthogonal to the cylinder center line D of the switching valve seat element **25** (or the valve seat cylindrical portion **62**), and is opened in the outer

peripheral surface of the valve seat cylindrical portion **62**.

(159) As shown in FIG. **17(a)**, the valve seat disk **63** has a plate diameter equal to the outer diameter **D1** of the valve seat cylindrical portion **62**, and is formed into a circular shape. The valve seat disk **63** is arranged concentrically with the valve seat cylindrical portion **62** with the cylinder center line **D** of the switching valve seat element **25** (or the valve seat cylindrical portion **62**) being a center. The valve seat disk **63** is formed integrally with the valve seat cylindrical portion **62** so as to close one cylinder end **62A** of the valve seat cylindrical portion **62**.

(160) As shown in FIG. **17(a)**, the valve seat holes **64** and **65** are each a circular hole having a hole diameter **d4**, and are formed in the valve seat disk **63**. As shown in FIG. **17(a)**, the valve seat holes **64** and **65** are arranged (in a concyclic manner) on a circle **CA** having a circle diameter **D5** and a center along the cylinder center line **D** of the switching valve seat element **25**. Each of the valve seat holes **64** and **65** is arranged so that a hole center line **E** thereof is located on the circle **CA**.

(161) As shown in FIG. **16(a)** and FIG. **17**, the valve seat holes **64** and **65** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching valve seat element **25** (or the valve seat cylindrical portion **62**).

(162) The valve seat holes **64** and **65** are each formed to pass through the valve seat disk **63** in a direction of the cylinder center line **D** of the switching valve seat element **25**, and are each opened in a disk front flat surface **63A** and a disk back flat surface **63B** of the valve seat disk **63**. The valve seat holes **64** and **65** communicate with an inside of the valve seat cylindrical portion **62**.

(163) As shown in FIG. **16**, FIG. **17(b)**, and FIG. **18(b)**, the first regulating protrusions **66** are arranged (in a concyclic manner) on a circle that has a center along the cylinder center line **D** of the switching valve seat element **25** (or the valve seat cylindrical portion **62**) and is located between the valve seat hole **64** and the outer peripheral surface of the valve seat cylindrical portion **62**. The first regulating protrusions **66** are formed integrally with the other cylinder end **62B** of the valve seat cylindrical portion **62** so as to be located on the valve seat hole **64** side.

(164) The first regulating protrusions **66** are arranged on both sides of a valve seat straight line **LB** that is orthogonal to the cylinder center line **D** of the switching valve seat element **25** and passes the hole center line **E** of each of the valve seat holes **64** and **65**. As shown in FIG. **17(b)**, each of the first regulating protrusions **66** is arranged at a distance **HC/2** from the valve seat straight line **LB**.

(165) This configuration, as shown in FIG. **17(b)**, allows the first regulating protrusions **66** to be arranged with an insertion interval **HC** in the peripheral direction of the switching valve seat element **25**. The insertion interval **HC** is larger than the rib width **hA** of each of the first rib portions **50** (of the switching base **22**) and smaller than the rib width **hB** of each of the second rib portions **51** (the rib width **hA** < the insertion interval **HC** < the rib width **hB**).

(166) In the direction of the cylinder center line **D** of the switching valve seat element **25**, the first regulating protrusions **66** are each formed to protrude from the other cylinder end **62B** of the valve seat cylindrical portion **62** and extend away from the valve seat disk **63**.

(167) As shown in FIG. **17(b)** and FIG. **18(a)**, the second regulating protrusions **67** are arranged on the same circle on which the first regulating protrusions **66** are arranged. Each of the second regulating protrusions **67** is arranged with an angular interval of 180 degrees in the peripheral direction of the switching valve seat element **25** with respect to one of the first regulating protrusions **66**, and are located on the valve seat hole **65** side.

(168) The second regulating protrusions **67** are arranged on the both sides of the valve seat straight line **LB**. Each of the second regulating protrusions **67** is arranged at the distance **HC/2** from the valve seat straight line.

(169) This configuration allows the second regulating protrusions **67** to be arranged with the insertion interval **HC** in the peripheral direction of the switching valve seat element **25**.

(170) In the direction of the cylinder center line **D** of the switching valve seat element **25**, the second regulating protrusions **67** are each formed to protrude from the other cylinder end **62B** of the valve seat cylindrical portion **62** and extend away from the valve seat disk **63**.

(171) As shown in FIG. 16(b), FIG. 17(b), and FIG. 18(b), the spring receiving protruding portions **68** are located in the valve seat cylindrical portion **62**, and are arranged between the valve seat holes **64** and **65**. The spring receiving protruding portions **68** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching valve seat element **25**. The spring receiving protruding portions **68** are arranged concentrically with the valve seat cylindrical portion **62** with the cylinder center line D of the switching valve seat element **25** being a center. As shown in FIG. 17(b), the spring receiving protruding portions **68** are each formed into an arc shape so as to have a radius r2 from the cylinder center line D (the center line) of the switching valve seat element **25**. The radius r2 of each of the spring receiving protruding portions **68** is smaller than a distance (length) between the cylinder center line D of the switching valve seat element **25** and the valve seat hole **64**.

(172) The spring receiving protruding portions **68** are formed integrally with the valve seat disk **63**. The spring receiving protruding portions **68** are formed to protrude from the disk back flat surface **63B** of the valve seat disk **63** into the valve seat cylindrical portion **62** in the direction of the cylinder center line D of the switching valve seat element **25**.

(173) As shown in FIG. 4 and FIG. 18, the sealing ring **26** is made of an elastic material such as synthetic rubber, and is formed into an annular shape. The sealing ring **26** is externally fitted to the valve seat cylindrical portion **62** of the switching valve seat element **25**, and is fitted in the sealing groove **69**. The sealing ring **26** is arranged in the sealing groove **69** so as to protrude from the outer peripheral surface of the valve seat cylindrical portion **62**.

(174) As shown in FIG. 19 to FIG. 25, the switching valve element **27** is made of a synthetic resin and formed into a cylindrical shape. The switching valve element **27** includes a first valve element cylindrical portion **71** (a large-diameter cylindrical portion), a valve element annular plate **72**, a second valve element cylindrical portion **73** (a small-diameter cylindrical portion), a valve element disk **74**, a central cylindrical portion **75**, a plurality of (a pair of) cylindrical valve elements **76** and **77**, a plurality of (a pair of) valve element flow passages **78** and **79**, a plurality of (a pair of) first valve element protruding portions **80**, a plurality of (a pair of) second valve element protruding portions **81**, a plurality of outer outflow holes **82**, a plurality of (a pair of) first handle regulating protruding portions **83**, and a plurality of (a pair of) second handle regulating protruding portions **85**.

(175) As shown in FIG. 19 to FIG. 25, the first valve element cylindrical portion **71** is formed into a cylindrical shape. As shown in FIG. 10 and FIG. 20, an outer diameter D2 of the first valve element cylindrical portion **71** is smaller than a hole diameter d2 of the medium-diameter hole portion **33B** of the handle hole **33** (of the switching handle **21**) (the outer diameter D2 < the hole diameter d2). As shown in FIG. 17(a) and FIG. 23, an inner diameter d3 of the first valve element cylindrical portion **71** is larger than the outer diameter D1 of the valve seat cylindrical portion **62** and the valve seat disk **63** (of the switching valve seat element **25**) (inner diameter d3 > outer diameter D1).

(176) As shown in FIG. 19 to FIG. 25, the valve element annular plate **72** is formed into an annular shape. The valve element annular plate **72** has the outer diameter D2 equal to the outer diameter of the first valve element cylindrical portion **71**.

(177) The valve element annular plate **72** is arranged concentrically with the first valve element cylindrical portion **71** with a cylinder center line F (a center line) of the switching valve element **27** (or the first valve element cylindrical portion **71**) being a center. The valve element annular plate **72** is formed integrally with the first valve element cylindrical portion **71** so as to close one cylinder end **71A** of the first valve element cylindrical portion **71**.

(178) As shown in FIG. 19 to FIG. 24, the second valve element cylindrical portion **73** is arranged concentrically with the first valve element cylindrical portion **71** with the cylinder center line F of the switching valve element **27** (or the first valve element cylindrical portion **71**) being a center. The second valve element cylindrical portion **73** is arranged along an inner periphery of the valve

element annular plate **72**, and is formed integrally with the valve element annular plate **72**.

(179) The second valve element cylindrical portion **73** is formed to protrude from the valve element annular plate **72** in a direction of the cylinder center line F of the switching valve element **27**. An outer diameter D3 of the second valve element cylindrical portion **73** is smaller than the inner diameter d3 of the first valve element cylindrical portion **71** (the outer diameter D3 < the inner diameter d3).

(180) The second valve element cylindrical portion **73** has a shower outflow hole **87**. The shower outflow hole **87** is arranged concentrically with the second valve element cylindrical portion **73** with the cylinder center line F of the switching valve element **27** being a center. The shower outflow hole **87** is formed to pass through the second valve element cylindrical portion **73** in the direction of the cylinder center line F of the switching valve element **27** (or the first valve element cylindrical portion **71**). The shower outflow hole **87** is opened to one cylinder end **73A** and the other cylinder end **73B** of the second valve element cylindrical portion **73**.

(181) As shown in FIG. **19(a)**, FIG. **20**, FIG. **23**, and FIG. **24**, the shower outflow hole **87** includes a large-diameter hole portion **87A** and a small-diameter hole portion **87B**. The large-diameter hole portion **87A** is opened to the protruding-side cylinder end **73A** (one cylinder end) of the second valve element cylindrical portion **73**. The small-diameter hole portion **87B** is reduced in diameter at a hole step portion **87C** as compared to the large-diameter hole portion **87A**, and is opened to the other cylinder end **73B** of the second valve element cylindrical portion **73**.

(182) As shown in FIG. **19** to FIG. **21** and FIG. **23** to FIG. **25**, the valve element disk **74** is formed into a circular shape. The valve element disk **74** is arranged concentrically with the second valve element cylindrical portion **73** with the cylinder center line F of the switching valve element **27** being a center. The valve element disk **74** is arranged in the small-diameter hole portion **83B** of the second valve element cylindrical portion **73** to close the other cylinder end **73B** of the second valve element cylindrical portion **73**. The valve element disk **74** is formed integrally with the second valve element cylindrical portion **73**.

(183) As shown in FIG. **19(b)**, FIG. **20**, and FIG. **23** to FIG. **25**, the central cylindrical portion **75** is arranged concentrically with the valve element cylindrical portions **71** and **73** with the cylinder center line F of the switching valve element **27** being a center. The central cylindrical portion **75** is arranged in the second valve element cylindrical portion **73** (or in the shower outflow hole **87**). In a direction orthogonal to the cylinder center line F of the switching valve element **27**, the central cylindrical portion **75** is arranged at a center of each of the valve element cylindrical portions **71** and **73** with an annular space between an inner peripheral surface of the second valve element cylindrical portion **73** and the central cylindrical portion **75**.

(184) As shown in FIG. **23** and FIG. **24**, the central cylindrical portion **75** is formed integrally with the valve element disk **74** so that one cylinder end **75A** of the central cylindrical portion **75** is fixed to a disk back flat surface **74B** of the valve element disk **74**. The central cylindrical portion **75** is formed to extend from the disk back flat surface **74B** of the valve element disk **74** into the first valve element cylindrical portion **71** in the direction of the cylinder center line F of the switching valve element **27**. The central cylindrical portion **75** is formed to protrude from the first valve element cylindrical portion **71** in the direction of the cylinder center line F of the switching valve element **27**.

(185) As shown in FIG. **19(b)** and FIG. **21** to FIG. **24**, the cylindrical valve elements **76** and **77** are each formed into a cylindrical shape. The cylindrical valve elements **76** and **77** are arranged in the second valve element cylindrical portion **73** (or in the first valve element cylindrical portion **71**).

(186) As shown in FIG. **21(a)**, the cylindrical valve elements **76** and **77** are arranged (in a concyclic manner) on a circle CB that has a circle diameter D6 and a center along the cylinder center line F of the switching valve element **27** (or the first valve element cylindrical portion **71**), and is located between the central cylindrical portion **75** and the second valve element cylindrical portion **73**. The cylindrical valve elements **76** and **77** are arranged adjacent to the central cylindrical portion **75** so

that a cylinder center line G of each of the cylindrical valve elements **76** and **77** is located on the circle CB. The circle diameter D6 of the circle CB, on which the cylindrical valve elements **76** and **77** are arranged, is equal to the circle diameter D5 of the circle CA, on which the valve seat holes **64** and **65** are arranged (the circle diameter D6=the circle diameter D5).

(187) The cylindrical valve elements **76** and **77** are formed integrally with the central cylindrical portion **75**.

(188) The cylindrical valve elements **76** and **77** are formed integrally with the valve element disk **74** so as to be fixed to the disk back flat surface **74B** of the valve element disk **74**. The cylindrical valve elements **76** and **77** are each formed to extend from the disk back flat surface **74B** of the valve element disk **74** into the first valve element cylindrical portion **71** in the direction of the cylinder center line F of the switching valve element **27** (or the first valve element cylindrical portion **71**). The cylindrical valve elements **76** and **77** are each formed to protrude from the first valve element cylindrical portion **71** in the direction of the cylinder center line F of the switching valve element **27**.

(189) Cylinder ends **76A** and **77A** of the cylindrical valve elements **76** and **77** and a cylinder end **75A** of the central cylindrical portion **75**, which protrude from the first valve element cylindrical portion **71**, are formed into flat end surfaces that are flush with each other.

(190) As shown in FIG. **19(b)**, FIG. **20**, FIG. **21(a)**, and FIG. **24**, the cylindrical valve element **76** has a valve element hole **88** and a sealing groove **89**.

(191) As shown in FIG. **19(b)**, FIG. **20**, FIG. **21(a)**, FIG. **24**, and FIG. **25**, the valve element hole **88** is formed into a circular hole having a hole diameter d5. The valve element hole **88** is arranged concentrically with the cylindrical valve element **76** with the cylinder center line G of the cylindrical valve element **76** being a center. The valve element hole **88** is formed to extend from one cylinder end **76A** of the cylindrical valve element **76** to the valve element disk **74** in a direction of the cylinder center line G (the center line) of the cylindrical valve element **76**, and is opened to the one cylinder end **76A** of the cylindrical valve element **76**. The valve element hole **88** is closed by the valve element disk **84** in the direction of the cylinder center line G of the cylindrical valve element **76**.

(192) The hole diameter d5 of the valve element hole **88** is larger than the hole diameter d4 of each of the valve seat holes **64** and **65** (the hole diameter d5<the hole diameter d4).

(193) As shown in FIG. **19(b)** and FIG. **21(a)**, the sealing groove **89** is an annular groove, and is formed on the one cylinder end **76A** side of the cylindrical valve element **76**. The sealing groove **89** is arranged concentrically with the cylindrical valve element **76** with the cylinder center line G of the cylindrical valve element **76** being a center. The sealing groove **89** is arranged on an outer side of the valve element hole **88** in a direction orthogonal to the cylinder center line G of the cylindrical valve element **76**. The sealing groove **89** has a groove depth in the direction of the cylinder center line G of the cylindrical valve element **76**, and is opened to the one cylinder end **76A** of the cylindrical valve element **76**.

(194) As shown in FIG. **19(b)**, FIG. **20**, FIG. **21(a)**, FIG. **24**, and FIG. **25**, the cylindrical valve element **77** has a valve element hole **90** and a sealing groove **91**.

(195) As shown in FIG. **19(b)**, FIG. **20**, FIG. **21(a)**, FIG. **24**, and FIG. **25**, the valve element hole **90** is formed into a circular hole having the hole diameter d5. The valve element hole **90** is arranged concentrically with the cylindrical valve element **77** with the cylinder center line G of the cylindrical valve element **77** being a center. The valve element hole **90** is formed to extend from one cylinder end **77A** of the cylindrical valve element **77** to the valve element disk **74** in a direction of the cylinder center line G (the center line) of the cylindrical valve element **77**, and is opened to the one cylinder end **77A** of the cylindrical valve element **77**. The valve element hole **90** is closed by the valve element disk **74** in the direction of the cylinder center line G of the cylindrical valve element **77**.

(196) As shown in FIG. **19(a)** and FIG. **21(a)**, the sealing groove **91** is an annular groove, and is

formed on the one cylinder end 77A side of the cylindrical valve element 77. The sealing groove 91 is arranged concentrically with the cylindrical valve element 77 with the cylinder center line G of the cylindrical valve element 77 being a center. The sealing groove 91 is arranged on an outer side of the valve element hole 90 in the direction orthogonal to the cylinder center line G of the cylindrical valve element 77. The sealing groove 91 has a groove depth in the direction of the cylinder center line G of the cylindrical valve element 77, and is opened to the one cylinder end 77A of the cylindrical valve element 77.

(197) As shown in FIG. 19(a), FIG. 20, FIG. 21(a), and FIG. 22 to FIG. 25, the valve element flow passage 78 is formed in the valve element disk 74 within the small-diameter hole portion 87B of the shower outflow hole 87. As shown in FIG. 20, on a valve element transverse straight line LC that is orthogonal to the cylinder center line F of the switching valve element 27 and passes the cylinder center line G of each of the cylindrical valve elements 76 and 77, the valve element flow passage 78 is formed in one of halves of the valve element disk 74 (the upper half of the valve element disk 74) divided by the valve element transverse straight line LC.

(198) The valve element flow passage 78 is opened inside of the valve element hole 88 on the one cylinder end 76A side of the cylindrical valve element 76. The valve element flow passage 78 is formed to extend helically along an outer peripheral surface of the central cylindrical portion 75 while inclining toward a disk front flat surface 74A of the valve element disk 74 from a portion on the one cylinder end 76A side of the cylindrical valve element 76 on which the valve element flow passage 78 is opened inside of the valve element hole 88.

(199) The valve element flow passage 78 is formed to extend to a position above the cylindrical valve element 77 (or above the valve element hole 90) and with an angular interval of 90 degrees from the portion on the one cylinder end 76A side of the cylindrical valve element 76, on which the valve element flow passage 78 is opened inside of the valve element hole 88, in the peripheral direction of the switching valve element 27. The valve element flow passage 78 is located at the position above the cylindrical valve element 77 in the disk front flat surface 74A of the valve element disk 74.

(200) The valve element flow passage 78 is opened in the disk front flat surface 74A of the valve element disk 74 between the portion on the one cylinder end 76A side of the cylindrical valve element 76 and the cylindrical valve element 77, and communicates with the small-diameter hole portion 87B of the shower outflow hole 87.

(201) In the upper half of the valve element disk 74, the valve element flow passage 78 is formed by recessing (or protruding) a portion of the valve element disk 74 adjacent to the central cylindrical portion 75 into a helical shape toward the one cylinder end 76A side of the cylindrical valve element 76 along the outer peripheral surface of the central cylindrical portion 75.

(202) Thus, the valve element flow passage 78 is formed into a helical flow passage extending from the portion on the one cylinder end 76A side of the cylindrical valve element 76 to the position above the cylindrical valve element 77 (or above the valve element hole 90) along the outer peripheral surface of the central cylindrical portion 75.

(203) As shown in FIG. 19(a), FIG. 20, FIG. 21(a), and FIG. 22 to FIG. 25, the valve element flow passage 79 is formed in the valve element disk 74 within the small-diameter hole portion 87B of the shower outflow hole 87. As shown in FIG. 20, the valve element flow passage 79 is formed in the other one of halves of the valve element disk 74 (the lower half of the valve element disk 74) divided by the valve element transverse straight line LC.

(204) The valve element flow passage 79 is opened inside of the valve element hole 90 on the one cylinder end 77A side of the cylindrical valve element 77. The valve element flow passage 79 is formed to extend helically along the outer peripheral surface of the central cylindrical portion 75 while inclining toward the disk front flat surface 74A of the valve element disk 74 from a portion on the one cylinder end 77A side of the cylindrical valve element 77 on which the valve element flow passage 79 is opened inside of the valve element hole 90.

(205) The valve element flow passage **79** is formed to extend to a position above the cylindrical valve element **76** (or above the valve element hole **88**) and with an angular interval of 90 degrees from the portion on the one cylinder end **77A** side of the cylindrical valve element **77**, on which the valve element flow passage **79** is opened inside of the valve element hole **90**, in the peripheral direction of the switching valve element **27**. The valve element flow passage **79** is located at the position above the cylindrical valve element **76** in the disk front flat surface **74A** of the valve element disk **74**.

(206) The valve element flow passage **79** is opened in the disk front flat surface **74A** of the valve element disk **74** between the portion on the one cylinder end **77A** side of the cylindrical valve element **77** and the cylindrical valve element **76**, and communicates with the small-diameter hole portion **87B** of the shower outflow hole **87**.

(207) In the lower half of the valve element disk **74**, the valve element flow passage **79** is formed by recessing (or protruding) a portion of the valve element disk **74** adjacent to the central cylindrical portion **75** into a helical shape toward the one cylinder end **77A** side of the cylindrical valve element **77** along the outer peripheral surface of the central cylindrical portion **75**.

(208) Thus, the valve element flow passage **79** is formed into a helical flow passage extending from the portion on the one cylinder end **77A** side of the cylindrical valve element **77** to the position above the cylindrical valve element **76** (or above the valve element hole **88**) along the outer peripheral surface of the central cylindrical portion **75**.

(209) As shown in FIG. **19** to FIG. **22**, FIG. **24**, and FIG. **25**, the first valve element protruding portions **80** are formed on the first valve element cylindrical portion **71**. The valve element protruding portions **80** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching valve element **27** on the valve element transverse straight line LC. The first valve element protruding portions **80** are each formed to protrude from an outer peripheral surface of the first valve element cylindrical portion **71** in a direction orthogonal to the cylinder center line F (the center line) of the switching valve element **27** (or in the direction of the valve element transverse straight line LC). A protruding amount of each of the first valve element protruding portions **80** is set smaller than the groove depth of each of the first retaining grooves **35** (of the switching handle **21**).

(210) The first valve element protruding portions **80** are each formed to have a width $hC/2$ on each side thereof with respect to the valve element longitudinal straight line LC, and have a protruding width hC in the peripheral direction of the switching valve element **27**.

(211) The protruding width hC of each of the first valve element protruding portions **80** is set smaller than the groove width hA of each of the first retaining grooves **35** (of the switching handle **21**).

(212) As shown in FIG. **19**, FIG. **22**, and FIG. **24**, the first valve element protruding portions **80** are each formed to extend from the first valve element cylindrical portion **71** to the one cylinder end **76A** side of the cylindrical valve element **76** and the one cylinder end **77A** side of the cylindrical valve element **77** in the direction of the cylinder center line F of the switching valve element **27**.

(213) As shown in FIG. **19** to FIG. **23** and FIG. **25**, the second valve element protruding portions **81** are formed on the first valve element cylindrical portion **71**. The second valve element protruding portions **81** are arranged with an angular interval of 180 degrees in the peripheral direction of the switching valve element **27**. The second valve element protruding portions **81** are arranged on a valve element longitudinal straight line LD that is orthogonal to the cylinder center line F of the switching valve element **27** and the valve element transverse straight line LC. The second valve element protruding portions **81** are each formed to protrude from the outer peripheral surface of the first valve element cylindrical portion **71** in the direction orthogonal to the cylinder center line F of the switching valve element **27** (or in a direction of the valve element longitudinal straight line LD). A protruding amount of each of the second valve element protruding portions **81** is set smaller than the groove depth of each of the second retaining grooves **36** (of the switching

handle **21**).

(214) The second valve element protruding portions **81** are each formed to have a width $hD/2$ on each side thereof with respect to the valve element longitudinal straight line LD, and have a protruding width hD in the peripheral direction of the switching valve element **27**. The protruding width hD of each of the second valve element protruding portions **81** is set smaller than the groove width hB of each of the second retaining grooves **36** (of the switching handle **21**).

(215) As shown in FIG. **19** to FIG. **21** and FIG. **23** to FIG. **25**, the plurality of outer outflow holes **82** are composed by, for example, forming twelve holes in the valve element annular plate **72**. The outer outflow holes **82** are arranged (in a concyclic manner) on a circle having a center along the cylinder center line F (the center line) of the switching valve element **27**. The outer outflow holes **82** are arranged at equal angular intervals (equal pitches), for example, with angular intervals of 30 degrees in the peripheral direction of the switching valve element **27**.

(216) The outer outflow holes **82** are formed to pass through the valve element annular plate **72** in the direction of the cylinder center line F of the switching valve element **27**, and are opened in a disk front flat surface **72A** and a disk back flat surface **72B** of the valve element annular plate **72**.

(217) This configuration allows the outer outflow holes **82** to communicate with an inside of the first valve element cylindrical portion **71** on an outer side of each of the cylindrical valve elements **76** and **77**.

(218) As shown in FIG. **19(b)**, FIG. **21(b)**, FIG. **22**, FIG. **24(b)**, and FIG. **25**, the first handle regulating protruding portions **83** are formed on the disk back flat surface **72B** of the valve element annular plate **72** and the disk back flat surface **74B** of the valve element disk **74**.

(219) The first handle regulating protruding portions **83** are formed to extend between the outer peripheral surface of the cylindrical valve element **76** and the inner peripheral surface of the first valve element cylindrical portion **71**, and are formed integrally with the cylindrical valve element **76** and the first valve element cylindrical portion **71**.

(220) The first handle regulating protruding portions **83** are arranged on both sides of the valve element transverse straight line LC in the peripheral direction of the switching valve element **27**. The first handle regulating protruding portions **83** each include a valve element regulating flat surface **83A** located at a valve element distance HD from the valve element transverse straight line LC. The valve element regulating flat surface **83A** is formed in parallel to the valve element transverse straight line LC. The valve element distance HD is equal to the base distance HA of the base protrusion **59** and the base distance HB of the base protrusion **60** (of the switching base **22**). The first handle regulating protruding portions **83** are each formed to protrude from the disk back flat surface **72B** of the valve element annular plate **72** and the disk back flat surface **74B** of the valve element disk **74** toward the one cylinder end **76A** side of the cylindrical valve element **76** in the direction of the cylinder center line F of the switching valve element **27**.

(221) As shown in FIG. **19(b)**, FIG. **21(b)**, FIG. **22(b)**, and FIG. **25**, the second handle regulating protruding portions **85** are formed on the disk back flat surface **72B** of the valve element annular plate **72** and the disk back flat surface **74B** of the valve element disk **74**.

(222) The second handle regulating protruding portions **85** are formed to extend between the outer peripheral surface of the cylindrical valve element **77** and the inner peripheral surface of the first valve element cylindrical portion **71**, and are formed integrally with the cylindrical valve element **77** and the first valve element cylindrical portion **71**.

(223) The second handle regulating protruding portions **85** are arranged on the both sides of the valve element transverse straight line LC in the peripheral direction of the switching valve element **27**. The second handle regulating protruding portions **85** each include a valve element regulating flat surface **85A** located at the valve element distance HD from the valve element transverse straight line LC. The valve element regulating flat surface **85A** is formed in parallel to the valve element transverse straight line LC.

(224) The second handle regulating protruding portions **85** are formed to protrude from the disk

back flat surface **72B** of the valve element annular plate **72** and the disk back flat surface **74B** of the valve element disk **74** toward the one cylinder end **77A** side of the cylindrical valve element **77** in the direction of the cylinder center line **F** of the switching valve element **27**.

(225) As shown in FIG. **4** and FIG. **24**, the sealing rings **28** are each made of an elastic material such as synthetic rubber, and are each formed into an annular shape.

(226) The sealing rings **28** are fitted in the sealing groove **89** of the cylindrical valve element **76** and the sealing groove **91** of the cylindrical valve element **77**, respectively. The sealing rings **28** are arranged in the sealing grooves **89** and **91** so as to protrude from the cylinder end **76A** of the cylindrical valve element **76** and the cylinder end **77A** of the cylindrical valve element **77**.

(227) As shown in FIG. **30** to FIG. **41**, the flow passage switching means **2** is accommodated (arranged) in the shower space **7C** and the outflow passage **10** (or the shower cylindrical portion **8**) of the shower main body **1**.

(228) In the flow passage switching means **2**, as shown in FIG. **26** to FIG. **29**, the switching base **22** is inserted into the switching handle **21** so that a handle unit **HU** is assembled.

(229) As shown in FIG. **26**, FIG. **27**, and FIG. **29**, the switching base **22** is inserted from the one cylinder end **46A** of the second base cylindrical portion **46** into the handle hole **33** (or into the large-diameter hole portion **33A**) of the switching handle **21**.

(230) The switching base **22** is arranged so that the base annular plate **47** is inserted in the medium-diameter hole portion **33B** of the switching handle **21**, and that the first base cylindrical portion **45** and the sealing gasket **23** are inserted in the small-diameter hole portion **33C** of the switching handle **21**. As shown in FIG. **26** to FIG. **29**, the switching base **22** is inserted in the handle hole **33** so that one of the first rib portions **50** and the base regulating groove **55** are arranged at positions corresponding to the first retaining groove **35** which is formed on the handle protrusion **37** of the switching handle **21**, to the handle protrusion **37**, and to the shower protruding portion **38**.

(231) The base annular plate **47** is brought into abutment against the second hole step portion **33E** of the switching handle **21** in the medium-diameter hole portion **33B** of the handle hole **33**, thereby placing the switching base **22** concentrically with the switching handle **21**.

(232) When the switching base **22** is placed in the switching handle **21**, the one cylinder end **46A** of the second base cylindrical portion **46** of the switching base **22** and the sealing ring **24** (or the sealing groove **54**) are arranged to protrude from the one cylinder end **31A** of the first handle cylindrical portion **31** of the switching handle **21**, and extend in the direction of the cylinder center line **B** of the switching handle **21**.

(233) Further, when the switching base **22** is placed in the switching handle **21**, as shown in FIG. **29**, the sealing gasket **23** is brought into press-contact with the inner peripheral surface of the small-diameter hole portion **33C** (or the handle hole **33**) of the switching handle **21**, thereby sealing the small-diameter hole portion **33C** of the handle hole **33** in a liquid-tight manner. Due to an elastic force of the sealing gasket **23**, a gap is defined between the outer peripheral surface of the base annular plate **47** of the switching base **22** and the medium-diameter hole portion **33B** of the switching handle **21**.

(234) Thus, the switching handle **21** is freely turn around the switching base **22**.

(235) The switching handle **21** is turned under a state in which the small-diameter hole portion **33C** of the handle hole **33** is in sliding contact with the sealing gasket **23** in the switching base **22**. As shown in FIG. **29**, the large-diameter hole portion **33A** (or the handle hole **33**) of the switching handle **21** communicates with the base inflow passages **Z** through the small-diameter hole portion **48A** (or the base hole **48**) of the switching base **22**.

(236) As shown in FIG. **26** and FIG. **29**, the base protrusions **59** and **60** of the switching base **22** are arranged to protrude in the medium-diameter hole portion **33B** (or in the handle hole **33**) of the switching handle **21**.

(237) Thus, in the flow passage switching means **2**, the switching base **22** is placed in the switching handle **21** so that the handle unit **HU** is assembled.

(238) In the flow passage switching means 2, as shown in FIG. 30 to FIG. 32, the handle unit HU (including the switching handle 21 and the switching base 22) is arranged in the shower space 7C and the outflow passage 10 (or in the shower cylindrical portion 8) of the shower main body 1. As shown in FIG. 30, the handle unit HU is inserted from the second base cylindrical portion 46 of the switching base 22 into the shower space 7C and the outflow passage 10 of the shower main body 1 (or the head portion 7). The handle unit HU is arranged concentrically with the center line A of the outflow passage 10 (or the shower cylindrical portion 8).

(239) As shown in FIG. 30 to FIG. 32, the handle unit HU is inserted in the shower main body 1 so that the shower protruding portion 38, the handle protrusion 37, one of first retaining grooves 35 of the switching handle 21 and the base regulating groove 55 of the switching base 22 are arranged at positions corresponding to the reference protruding portion 14 (or the highest point 7a) of the head portion 7.

(240) With regard to the handle unit HU, the second base cylindrical portion 46 of the switching base 22 is inserted into the shower cylindrical portion 8 (or into the outflow passage 10) from the cylinder end 46A, and the first handle cylindrical portion 31 of the switching handle 21 is inserted into the guide protruding portion 12 and the shower space 7C.

(241) In the handle unit HU, as shown in FIG. 32, the second base cylindrical portion 46 of the switching base 22 is accommodated in the shower cylindrical portion 8 (or in the outflow passage 10) so that the fixing protruding portions 11 of the shower main body 1 are inserted in the base regulating grooves 55, 56, and 57, respectively.

(242) Thus, the switching base 22 is mounted to the head portion 7 of the shower main body 1 so as to be unturnable.

(243) As shown in FIG. 30 to FIG. 32, the first rib portions 50 of the switching base 22 are arranged at positions corresponding to the reference protruding portion 14 of the shower main body 1.

(244) In the handle unit HU, the second base cylindrical portion 46 of the switching base 22 is inserted in the outflow passage 10 under a state in which the sealing ring 24 is held in press-contact with the inner peripheral surface of the shower cylindrical portion 8 (or the outflow passage 10). The handle unit HU is placed on the handle portion 6 under a state in which the one cylinder end 46A of the second base cylindrical portion 46 is held in abutment against the hole step portion 10C of the outflow passage 10.

(245) In the handle unit HU, as shown in FIG. 30, the fixing cylindrical portion 49 of the switching base 22 is inserted in the outflow passage 10 (or in the shower cylindrical portion 8), and is arranged under a state in which the base protruding portion 13 of the shower main body 1 is press-fitted in the medium-diameter hole portion 58C of the bolt receiving hole 58.

(246) Thus, the bolt receiving hole 58 of the switching base 22 communicates with the screw hole 15 of the base protruding portion 13.

(247) In the handle unit HU, as shown in FIG. 30, the first handle cylindrical portion 31 of the switching handle 21 is inserted in the guide protrusion 12 and the shower space 7C of the shower main body 1. The switching handle 21 is arranged under a state in which the guide protruding portion 12 of the shower main body 1 is inserted in the handle groove 40. The guide protrusion 12 of the shower main body 1 is inserted into the handle groove 40 without contact with the switching handle 21. The switching handle 21 is arranged under a state in which the protrusion end surface 37A of the handle protrusion 37 is held in abutment against the one cylinder end 8A of the shower cylindrical portion 8.

(248) When the handle unit HU is thus arranged in the shower space 7C and the outflow passage 10 of the shower main body 1, as shown in FIG. 30 to FIG. 32, the base inflow passages Z of the switching base 22 communicate with the outflow passage 10 on the dome top 7A side of the head portion 7, and communicate with the inflow passage 9 of the handle portion 6 through the outflow passage 10.

(249) In the handle unit HU, as shown in FIG. 30 to FIG. 32, the medium-diameter hole portion 33B of the switching handle 21 communicates with the outflow passage 10 through the base inflow passages Z and the small-diameter hole portion 48A (or the base hole 48) of the switching base 22.

(250) In the flow passage switching means 2, as shown in FIG. 33 and FIG. 34, when the handle unit HU (including the switching handle 21 and the switching base 22) is arranged in the shower main body 1 (or in the shower space 7C and the outflow passage 10), the switching base 22 is fixed to the shower main body 1 (or the head portion 7) with the fixing screw bolt 29.

(251) As shown in FIG. 33 and FIG. 34, the fixing screw bolt 29 is inserted in the fixing cylindrical portion 49 of the switching base 22.

(252) A bolt shank 29A is inserted into the large-diameter hole portion 58A and the small-diameter hole portion 58B (or the bolt receiving hole 58) of the fixing cylindrical portion 49 so that the fixing screw bolt 29 is screwed into the screw hole 15 of the base protruding portion 13 (or the shower main body 1). A bolt head 29B is inserted into the large-diameter hole portion 58A of the fixing cylindrical portion 49 so that the fixing screw bolt 29 is arranged to be held in abutment against the hole step portion 58D. Through turning of the fixing screw bolt 29, the second base cylindrical portion 46 of the switching base 22 is fastened to the base protruding portion 13. Thus, as shown in FIG. 33, the switching base 22 of the handle unit HU is fixed to the shower main body 1 (or the head portion 7) with the fixing screw bolt 29.

(253) The switching handle 21 of the handle unit HU is mounted to the shower main body so as to be freely turnable.

(254) In the switching base 22 of the handle unit HU, as shown in FIG. 34, the first base regulating flat surface 59A of the base protrusion 59 is arranged at the base distance HA from the shower protruding portion 38 of the shower main body 1.

(255) As shown in FIG. 33 and FIG. 34, when the switching base 22 of the handle unit HU of the flow passage switching means 2 is fixed to the shower main body 1 with the fixing screw bolt 29, the coil spring 30 is arranged in the switching base 22.

(256) As shown in FIG. 33 and FIG. 34, the coil spring 30 is arranged concentrically with the center line A of the outflow passage 10, and is inserted in the switching base 22. The coil spring 30 is inserted in the large-diameter hole portion 58A of the bolt receiving hole 58 of the fixing cylindrical portion 49 (of the switching base 22). The coil spring 30 is externally fitted to the bolt head 29B of the fixing screw bolt 29, and is inserted in the large-diameter hole portion 58A of the bolt receiving hole 58. The coil spring 30 is arranged so that one spring end thereof is held in abutment against the hole step portion 58D of the bolt receiving hole 58.

(257) Thus, as shown in FIG. 33 and FIG. 34, in the direction of the center line A of the outflow passage 10 (or the cylinder center line B of the switching handle 21), the coil spring 30 is arranged to protrude from the hole step portion 58D of the fixing cylindrical portion 49 into the small-diameter hole portion 48A (or into the base hole 48) of the switching base 22.

(258) In the flow passage switching means 2, as shown in FIG. 35 to FIG. 37, the switching valve seat element 25 is accommodated (arranged) in the handle unit HU (or the switching base 22) arranged in the shower main body 1.

(259) As shown in FIG. 35 to FIG. 37, the switching valve seat element 25 is arranged concentrically with the cylinder center line C of the switching base 22, and is inserted into the small-diameter hole portion 48A (or into the base hole 48) of the switching base 22 from the first regulating protrusions 66 and the second regulating protrusions 67.

(260) The switching valve seat element 25 is inserted in the small-diameter hole portion 48A of the switching base 22 under a state in which the first rib portions 50 of the switching base are located between the first regulating protrusions 66 (within the base distance HA) and between the second regulating protrusions 67 (within the base distance HA).

(261) As shown in FIG. 35 to FIG. 37, the switching valve seat element 25 is arranged in the switching base 22 under a state in which the valve seat disk 63 and the valve seat cylindrical

portion **61** are inserted in the small-diameter hole portion **48A** (or in the base hole **48**) of the switching base **22**. At this time, the sealing ring **26** on the switching valve seat element **25** (or the valve seat cylindrical portion **62**) is held in press-contact with the inner peripheral surface of the small-diameter hole portion **48A** of the base hole **48**, thereby sealing the small-diameter hole portion **48A** in a liquid-tight manner.

(262) As shown in FIG. **35** and FIG. **36**, the switching valve seat element **25** is inserted in the small-diameter hole portion **48A** of the switching base **22** under a state in which the other spring end side of the coil spring **30** is received within the spring receiving protruding portions **68** and the other spring end of the coil spring **30** is held in abutment against the disk back flat surface **63B** of the valve seat disk **63**.

(263) The switching valve seat element **25** is inserted in the small-diameter hole portion **48A** of the switching base **22** while compressing the coil spring **30**, which is received within the spring receiving protruding portions **68**, toward the switching base **22** side.

(264) As shown in FIG. **35** and FIG. **37**, the switching valve seat element **25** is arranged in the small-diameter hole portion **48A** of the switching base **22** under a state in which the first rib portion **50** of the switching base **22** is press-fitted between the first regulating protrusions **66** and the first rib portion **50** of the switching base **22** is press-fitted between the second regulating protrusions **67**.

(265) Thus, the switching valve seat element **25** is arranged in the switching base **22** and the shower main body **1** (or the head portion **7**) so as to be unturnable. The switching valve seat element **25** is freely movable in the direction of the cylinder center C of the switching base **22**.

(266) As shown in FIG. **5** to FIG. **37**, the valve seat holes **64** and **65** of the switching valve seat element **25** are arranged at positions corresponding to the reference protruding portion **14** of the shower main body **1** and the shower protruding portion **38** of the switching handle **21**, and communicate with the small-diameter hole portion **48A** of the switching base **22**.

(267) As shown in FIG. **36** and FIG. **37**, the valve seat holes **64** and **65** of the switching valve seat element **25** communicate with the outflow passage **10** and the inflow passage **9** through the base inflow passages Z of the switching base **22**.

(268) In the flow passage switching means **2**, as shown in FIG. **38** to FIG. **41**, the switching valve element **27** (switching valve) is arranged in the handle unit HU (or in the switching handle **21**) mounted to the shower main body **1**.

(269) As shown in FIG. **38** to FIG. **41**, the switching valve element **27** is arranged concentrically with the cylinder center line B of the switching handle **21**, and is inserted into the large-diameter hole portion **33A** and the medium-diameter hole portion **33B** (or into the handle hole **33**) of the switching handle **21** from the cylindrical valve elements **76** and **77** (or from the first handle regulating protruding portions **83** and the second handle regulating protruding portions **85**).

(270) As shown in FIG. **38** and FIG. **39**, the switching valve element **27** is arranged in the switching handle **21** of the handle unit HU under a state in which the first valve element cylindrical portion **71** is inserted in the medium-diameter hole portion **33B** (or in the handle hole **33**) of the switching handle **21**.

(271) As shown in FIG. **38**, FIG. **39**, and FIG. **41**, the switching valve element **27** is arranged in the switching handle **21** of the handle unit HU under a state in which the first valve element protruding portions **80** are respectively inserted in the first retaining grooves **35** of the switching handle **21** and the second valve element protruding portions **81** are respectively inserted in the second retaining grooves **36** of the switching handle **21**.

(272) Thus, the switching valve element **27** is mounted to the switching handle **21** so as to be unturnable, and is turned together with the switching handle **21**.

(273) As shown in FIG. **38** and FIG. **40**, the switching valve element **27** is arranged in the switching handle **21** under a state in which the cylindrical valve elements **76** and **77** are held in abutment against the disk front flat surface **63A** of the valve seat disk **63** of the switching valve seat element **25**. Each of the cylindrical valve elements **76** and **77** is held in abutment against the disk

front flat surface **63A** of the valve seat disk **63** through intermediation of the sealing ring **28**. As shown in FIG. **68**, the valve seat disk **63** of the switching valve seat element **25** is urged by a spring force of the coil spring **30** toward the sealing rings **28** in the cylindrical valve elements **76** and **77**. (274) As shown in FIG. **38** to FIG. **40**, the first valve element protruding portions **80** are respectively inserted in the first retaining grooves **35** of the switching handle **21** so that the cylindrical valve elements **76** and **77** of the switching valve element **27** are arranged at positions corresponding to the valve seat holes **64** and **65** of the switching valve seat element **25**, respectively.

(275) Thus, as shown in FIG. **68** and FIG. **70**, the valve element holes **88** and **90** of the cylindrical valve elements **76** and **77** of the switching valve element **27** are opened in the valve seat holes **64** and **65**, respectively.

(276) The cylindrical valve elements **76** and **77** (or the valve element holes **88** and **90**) communicate with the outflow passage **10** and the inflow passage **9** through the valve seat holes **64** and **65** of the switching valve seat element **25** and the base flow passages **Z** of the switching base **22**.

(277) As shown in FIG. **38**, FIG. **39**, and FIG. **41**, when the first valve element protruding portions **80** are inserted in the first retaining grooves **35** of the switching handle **21**, respectively, the switching valve element **27** is arranged so that the valve element regulating flat surface **83A** of one of the first handle regulating protruding portions **83** is held in abutment against the base protrusion **59** (or the first base regulating flat surface **59A**) of the switching base **22**, and that the valve element regulating flat surface **85A** of one of the second handle regulating protruding portions **85** is held in abutment against the base protrusion **60** (or the fourth base regulating flat surface **60B**) of the switching base **22**.

(278) Thus, as shown in FIG. **41**, the switching handle **21** and the switching valve element **27** are freely turnable between the base protrusions **59** and **60** of the switching base **22** within an angular range of 90 degrees.

(279) As shown in FIG. **38** and FIG. **39**, the second valve element cylindrical portion **73** of the switching valve element **27** is opened in the large-diameter hole portion **33A** of the switching handle **21**, and allows the valve element flow passages **78** and **79** (in the disk front flat surface **74A** of the valve element disk **74**) to communicate with an inside of the large-diameter hole portion **33A** (or an inside of the handle hole **33**) of the switching handle **21**.

(280) The valve element flow passages **78** and **79** of the switching valve element **27** communicate with the outflow passage **10** and the inflow passage **9** through the valve element holes **88** and **90**, the valve seat holes **64** and **65** of the switching valve seat element **25**, and the base flow passages **Z** of the switching base **22**.

(281) The valve element flow passages **78** and **79** communicate with the large-diameter hole portion **33A** (or the handle hole **33**) of the switching handle **21** through the shower outflow hole **87** of the second valve element cylindrical portion **73**.

(282) As shown in FIG. **38** and FIG. **39**, the outer outflow holes **82** of the switching valve element **27** are opened between the valve element annular plate **72** and the valve element disk **74** of the switching valve seat element **25**, and opened in the large-diameter hole portion **33A** (or in the handle hole **33**) of the switching handle **21**.

(283) With this configuration, the outer outflow holes **82** communicate with the outflow passage **10** and the inflow passage **9** through the valve seat holes **64** and **65** of the switching valve element **27** and the base inflow passages **Z** of the switching base **22**.

(284) Thus, as shown in FIG. **30** to FIG. **41**, the flow passage switching means **2** are arranged in the shower main body **1** (or in the head portion **7**), and is mounted to the shower main body **1**.

(285) In the shower head **X**, as shown in FIG. **1** to FIG. **4** and FIG. **42** to FIG. **45**, the shower nozzle **3** (spray nozzle) is mounted to the other end **1B** of the shower main body **1** (or the circular end **7B** of the head portion **7**).

(286) As shown in FIG. 42 to FIG. 45, the shower nozzle 3 is made of a synthetic resin and formed into a cylindrical shape. The shower nozzle 3 includes a nozzle outer cylindrical portion 95, a shower nozzle plate 96, a shower cylindrical portion 97 (nozzle inner cylindrical portion), a plurality of air bubble-liquid mixture jetting holes 98, and a sealing ring 103.

(287) As shown in FIG. 42, FIG. 44, and FIG. 45, the nozzle outer cylindrical portion 95 is formed into a cylindrical shape, and includes a sealing groove 99 and a threaded portion 100.

(288) As shown in FIG. 42 and FIG. 44, the sealing groove 99 is formed into an annular groove, and is arranged on one cylinder end 95A side of the nozzle outer cylindrical portion 95 in a direction of a cylinder center line H of the shower nozzle 3. The sealing groove 99 is arranged concentrically with the nozzle outer cylindrical portion 95 with the cylinder center line H (the center line) of the shower nozzle 3 (or the nozzle outer cylindrical portion 95) being a center. The sealing groove 99 is formed along an entire outer peripheral surface of the nozzle outer cylindrical portion 95. The sealing groove 99 has a groove depth in a direction orthogonal to the cylinder center line H of the shower nozzle 3, and is opened in the outer peripheral surface of the nozzle outer cylindrical portion 95.

(289) As shown in FIG. 42, FIG. 44, and FIG. 45, the threaded portion 100 is arranged on the other cylinder end 95B side of the nozzle outer cylindrical portion 95 in the direction of the cylinder center line H of the shower nozzle 3. The threaded portion 100 is arranged between the sealing groove 99 and the other cylinder end 95B of the nozzle outer cylindrical portion 95 in the direction of the cylinder center line H of the shower nozzle 3. The threaded portion 100 is formed along the entire outer peripheral surface of the nozzle outer cylindrical portion 95.

(290) As shown in FIG. 42 to FIG. 45, the shower nozzle plate 96 (spray nozzle plate) is formed into a circular plate. The shower nozzle plate 96 is arranged concentrically with the nozzle outer cylindrical portion 95 with the cylinder center line H of the shower nozzle 3 being a center.

(291) As shown in FIG. 43, the shower nozzle plate 96 has a plate diameter D7 equal to an outer diameter of the nozzle outer cylindrical portion 95, and closes the one cylinder end 95A of the nozzle outer cylindrical portion 95.

(292) The shower nozzle plate 96 is fixed to the one cylinder end 95A of the nozzle outer cylindrical portion 95, and is formed integrally with the nozzle outer cylindrical portion 95.

(293) As shown in FIG. 42(b), FIG. 44(b), and FIG. 45, the shower cylindrical portion 97 is formed into a cylindrical shape.

(294) The shower cylindrical portion 97 (spray cylindrical portion) is arranged concentrically with the nozzle outer cylindrical portion 95 and the shower nozzle plate 96 with the cylinder center line H of the shower nozzle 3 being a center. The shower cylindrical portion 97 is arranged in the nozzle outer cylindrical portion 95 with a mist annular space YM between an inner peripheral surface of the nozzle outer cylindrical portion 95 and the shower cylindrical portion 97 in the direction orthogonal to the cylinder center line H of the shower nozzle 3.

(295) One cylinder end 97A of the shower cylindrical portion 97 is closed by the shower nozzle plate 96, and the shower cylindrical portion 97 is formed integrally with the shower nozzle plate 96. The shower cylindrical portion 97 is formed to protrude from a disk back flat surface 96B of the shower nozzle plate 96 into the nozzle outer cylindrical portion 95 in the direction of the cylinder center line H of the shower nozzle 3.

(296) As shown in FIG. 42(b), FIG. 44(b), and FIG. 45, the shower cylindrical portion 97 is formed to be increased in diameter at a sealing step portion 101 on the shower nozzle plate 96 side. The sealing step portion 101 is formed into a circular shape, and is arranged concentrically with the shower cylindrical portion 97 with the cylinder center line H of the shower nozzle 3 being a center. The sealing step portion 101 is formed along an entire outer peripheral surface of the shower cylindrical portion 97.

(297) As shown in FIG. 42(b), FIG. 44(b), and FIG. 45, the shower cylindrical portion 97 has a nozzle hole 102.

(298) As shown in FIG. 44(b) and FIG. 45, the nozzle hole **102** is formed into a circular hole. The nozzle hole **102** is arranged concentrically with the shower cylindrical portion **97** with the cylinder center line H (the center line) of the shower nozzle **3** being a center. The nozzle hole **102** is formed to extend from the disk back flat surface **96B** of the shower nozzle plate **96** to the other cylinder end **97B** of the shower cylindrical portion **97** in the direction of the cylinder center line H of the shower nozzle **3**, and is opened to the other cylinder end **97B**.

(299) As shown in FIG. 42(b), FIG. 44(b), and FIG. 45, the nozzle hole **102** includes a large-diameter hole portion **102A**, a medium-diameter hole portion **102B**, and a small-diameter hole portion **102C**.

(300) The large-diameter hole portion **102A** is opened to the one cylinder end **97B** of the shower cylindrical portion **97**. The medium-diameter hole portion **102B** is arranged between the large-diameter hole portion **102A** and the small-diameter hole portion **102C**. The medium-diameter hole portion **102B** is reduced in diameter at a first hole step portion **102D** as compared to the large-diameter hole portion **102A**, and is formed to extend toward the shower nozzle plate **96** side. The small-diameter hole portion **102C** is reduced in diameter at a second hole step portion **102E** as compared to the medium-diameter hole portion **102B**, and is formed to extend to the shower nozzle plate **96** (or the disk back flat surface **96B**).

(301) With this configuration, the shower cylindrical portion **97** defines an air bubble mixing space BR into which the liquid flows from the other cylinder end **97B**. The air bubble mixing space BR is defined in the shower cylindrical portion **97** by the nozzle hole **102**.

(302) As shown in FIG. 44(b), the shower cylindrical portion **97** has the hole diameter d_5 at the small-diameter hole portion **102C** (or the nozzle hole **102**), and has a hole length L1 at the small-diameter hole portion **102C** in the direction of the cylinder center line H of the shower nozzle **3**.

(303) As shown in FIG. 42, FIG. 43, FIG. 44(b), and FIG. 45, the plurality of air bubble-liquid mixture jetting holes **98** are formed into circular throttle holes (nozzle throttle holes). Through the air bubble-liquid mixture jetting holes **98**, the air bubble-liquid mixture is jetted out of the air bubble mixing space BR.

(304) The air bubble-liquid mixture jetting holes **98** are formed in the shower nozzle plate **96**. The air bubble-liquid mixture jetting holes **98** are formed to pass through the shower nozzle plate **96** in the direction of the cylinder center line H of the shower nozzle **3**, and are opened into the air bubble mixing space BR in the shower cylindrical portion **97**.

(305) As shown in FIG. 43, the plurality of air bubble-liquid mixture jetting holes **98** are arranged (in a concyclic manner) on each of a plurality of circles CD, CE, and CF having different radii r_3 , r_4 , and r_5 ($r_3 < r_4 < r_5$) of the circles with the cylinder center line H (the center line) of the shower nozzle **3** being a center. On each of the circles CD, CE, and CF, the air bubble-liquid mixture jetting holes **98** are arranged at equal intervals (equal pitches) in the peripheral direction of the shower nozzle **3**.

(306) As shown in FIG. 44 and FIG. 45, the sealing ring **103** is made of an elastic material such as synthetic rubber, and is formed into an annular shape.

(307) The sealing ring **103** is externally fitted to the nozzle outer cylindrical portion **95**, and is fitted in the sealing groove **99**. The sealing ring **103** is arranged in the sealing groove **99** so as to protrude from the outer peripheral surface of the nozzle outer cylindrical portion **95**.

(308) In the shower head X, the air bubble-liquid mixture generating means **4** (the air bubble generating unit) is configured to generate the air bubble-liquid mixture by mixing the air (air bubbles) into the liquid.

(309) As shown in FIG. 2, FIG. 4, and FIG. 42 to FIG. 49, the air bubble-liquid mixture generating means **4** includes a flow-adjustment piece **111** and a plurality of (three) air introduction passages **112**.

(310) As shown in FIG. 46 to FIG. 49, the flow-adjustment piece **111** is made of a synthetic resin and formed into a cylindrical shape. The flow-adjustment piece **111** includes a flow-adjustment

cylindrical portion **113**, a flow-adjustment nozzle disk **114**, a flow-adjustment annular plate **115**, a plurality of (four) flow-adjustment-piece plates **116**, and a plurality of liquid throttle holes **117**.
(311) As shown in FIG. **46** to FIG. **49**, the flow-adjustment cylindrical portion **113** is formed into a cylindrical shape.

(312) As shown in FIG. **46** to FIG. **49**, the flow-adjustment nozzle disk **114** is a circular plate, and is formed to have a plate diameter equal to an outer diameter of the flow-adjustment cylindrical portion **113**. The flow-adjustment nozzle disk **114** is arranged concentrically with the flow-adjustment cylindrical portion **113** with a cylinder center line J (a center line) of the flow-adjustment piece **111** (or the flow-adjustment cylindrical portion **113**) being a center. The flow-adjustment nozzle disk **114** closes one cylinder end **113A** of the flow-adjustment cylindrical portion **113**, and is fixed to the flow-adjustment cylindrical portion **113**. The flow-adjustment nozzle disk **114** is formed integrally with the flow-adjustment cylindrical portion **113**.

(313) As shown in FIG. **46** to FIG. **49**, the flow-adjustment annular plate **115** is formed into an annular shape. The flow-adjustment annular plate **115** is arranged concentrically with the flow-adjustment cylindrical portion **113** and the flow-adjustment nozzle disk **114** with the cylinder center line J of the flow-adjustment piece **111** being a center. The flow-adjustment annular plate **115** is arranged on the other cylinder end **113B** side of the flow-adjustment cylindrical portion **113**.

(314) The flow-adjustment annular plate **115** is arranged at the other cylinder end **113B** of the flow-adjustment cylindrical portion **113** along an entire outer peripheral surface of the flow-adjustment cylindrical portion **113**, and is formed integrally with the flow-adjustment cylindrical portion **113**. The flow-adjustment annular plate **115** is formed to protrude from the outer peripheral surface of the flow-adjustment cylindrical portion **113** in a direction orthogonal to the cylinder center line J of the flow-adjustment piece **111** (or the flow-adjustment cylindrical portion **113**).

(315) As shown in FIG. **46** to FIG. **49**, the four flow-adjustment-piece plates **116** are formed on the flow-adjustment nozzle disk **114**.

(316) The flow-adjustment-piece plates **116** are each formed into a rectangular shape (rectangle). The flow-adjustment-piece plates **116** are arranged at equal angular intervals of 90 degrees in the circumferential direction of the flow-adjustment nozzle disk **114** (or the flow-adjustment piece **111**).

(317) The flow-adjustment-piece plates **116** are formed to protrude by a plate width HS from a disk front flat surface **114A** of the flow-adjustment nozzle disk **114** in a direction of the cylinder center line J (the center line) of the flow-adjustment piece **111**. The flow-adjustment-piece plates **116** are each formed to protrude in a direction orthogonal to the flow-adjustment nozzle disk **114** so as to be away from the other cylinder end **113B** of the flow-adjustment cylindrical portion **113**.

(318) As shown in FIG. **46(a)** and FIG. **47**, the flow-adjustment-piece plates **116** are each formed to extend by a plate length LS from the plate center line J of the flow-adjustment nozzle disk **114** (or the cylinder center line of the flow-adjustment piece **111**) toward the outer peripheral surface side of the flow-adjustment nozzle disk **114** (or the outer peripheral surface side of the flow-adjustment cylindrical portion **113**). The flow-adjustment-piece plates **116** are formed to extend in the direction orthogonal to the plate center line J of the flow-adjustment nozzle disk **114** with gaps along the outer peripheral surface of the flow-adjustment nozzle disk **114**. The flow-adjustment-piece plates **116** each have a plate thickness TS in the peripheral direction of the flow-adjustment nozzle disk **114** (or the peripheral direction of the flow-adjustment piece **111**).

(319) As shown in FIG. **46(a)**, FIG. **47**, FIG. **48**, and FIG. **49(b)**, the flow-adjustment-piece plates **116** each include flow-adjustment flat surfaces **116A** and **116B**, and a flow inclined surface **118**.

(320) The flow-adjustment flat surfaces **116A** and **116B** are each formed into a rectangular shape so as to be parallel to each other with an interval equal to the plate thickness TS in the peripheral direction of the flow-adjustment nozzle disk **114**.

(321) As shown in FIG. **48(b)**, in the direction of the cylinder center line J of the flow-adjustment piece **111**, the flow inclined surface **118** is formed to extend and incline from a protruding end

116D of each of the flow-adjustment-piece plates **116** (or one of plate ends) toward one flow-adjustment flat surface **116A** and the flow-adjustment nozzle disk **114** (or the disk front flat surface **114A**). For example, the flow inclined surface **118** is formed into an arc shape to protrude with a radius r_X between the protruding end **116D** of each of the flow-adjustment-piece plates **116** and the one flow-adjustment flat surface **116A**.

(322) As shown in FIG. **46**, FIG. **47**, and FIG. **49(a)**, the plurality of liquid throttle holes **117** are formed in the flow-adjustment nozzle disk **114** between the flow-adjustment-piece plates **116**. Each of the liquid throttle holes **117** is formed to pass through the flow-adjustment nozzle disk **114** in the direction of the cylinder center line J of the flow-adjustment piece **111** (or the plate center line J of the flow-adjustment nozzle disk **114**), and is opened in the disk front flat surface **114A** and a disk back flat surface **114B** of the flow-adjustment nozzle disk **114**. The liquid throttle holes **117** are formed to pass through the flow-adjustment nozzle disk **114** so that a hole center line M of each of the liquid throttle holes **117** is arranged in parallel to the plate center line J of the flow-adjustment nozzle disk **114**. The liquid throttle holes **117** are opened in the disk back flat surface **114B** of the flow-adjustment nozzle disk **114**, and communicate with an inside of the flow-adjustment cylindrical portion **113**.

(323) The liquid throttle holes **117** are each formed into a conical hole having a diameter gradually reducing from the disk back flat surface **114B** toward the disk front flat surface **114A** of the flow-adjustment nozzle disk **114** in the direction of the plate center line J of the flow-adjustment nozzle disk **114** (or the cylinder center line of the flow-adjustment piece **111**).

(324) As shown in FIG. **47**, the plurality of liquid throttle holes **117** are arranged on each of a plurality of circles CG, CH, and CI having different radii, r_6 , r_7 , and r_8 ($r_6 < r_6 < r_7$), of the circles with the plate center line J of the flow-adjustment nozzle disk **114** being a center.

(325) On each of the circles CG, CH, and CI, the plurality of liquid throttle holes **117** are arranged at equal intervals (equal pitches) in the peripheral direction (circumferential direction) of the flow-adjustment nozzle disk **114** (or the flow-adjustment piece **111**).

(326) As shown in FIG. **48(b)**, the flow-adjustment piece **111** has a piece height HP extending between the protruding end **116D** of each of the flow-adjustment-piece plates **116** and the other cylinder end **113B** of the flow-adjustment cylindrical portion **113** in the direction of the cylinder center line J of flow-adjustment piece **111**. The piece height HP is set smaller than the hole length L1 of the small-diameter hole portion **102C** of the shower cylindrical portion **97**.

(327) In the air bubble-liquid mixture generating means **4**, as shown in FIG. **42** to FIG. **45**, the plurality of (three) air introduction passages **112** are formed in the shower nozzle **3**.

(328) The air introduction passages **112** are arranged on a circle CJ that has a center along the cylinder center line H (the center line) of the shower nozzle **3** and is located on an outer side of the air bubble-liquid mixture jetting holes **98**. The air introduction passages **112** are arranged at equal angular intervals of 120 degrees in the peripheral direction of the shower nozzle **3** (or the shower cylindrical portion **97**).

(329) The air introduction passages **112** are opened in a disk front surface **96A** of the shower nozzle plate **96**. As shown in FIG. **44(b)**, the air introduction passages **112** are formed to extend from the disk front surface **96A** of the shower nozzle plate **96** toward the other cylinder end **97B** side of the shower cylindrical portion **97** in the direction of the cylinder center line H of the shower nozzle **3**. The air introduction passages **112** are formed on the cylinder end **97B** side of the shower cylindrical portion **97** to pass through the shower cylindrical portion **97** in the direction orthogonal to the cylinder center line H of the shower nozzle **3**.

(330) The air introduction passages **112** are opened into the air bubble mixing space BR in the shower cylindrical portion **97**. The air introduction passages **112** are adjacent to the second hole step portion **112E** of the shower cylindrical portion **97**, and are opened in the medium-diameter hole portion **102B** (or in the nozzle hole **102**).

(331) As shown in FIG. **50** and FIG. **51**, the flow-adjustment piece **111** of the air bubble-liquid

mixture generating means **4** is incorporated in the shower nozzle **3**.

(332) The flow-adjustment piece **111** is arranged concentrically with the shower cylindrical portion **97** with the cylinder center line H of the shower nozzle **3** being a center. The flow-adjustment piece **111** is arranged in the air bubble mixing space BR in the shower cylindrical portion **97**. The flow-adjustment piece **111** is press-fitted (inserted) into the nozzle hole **102** (or into the large-diameter hole portion **102A** and the medium-diameter hole portion **102B**) of the shower cylindrical portion **97** from the flow-adjustment-piece plates **116**.

(333) The flow-adjustment cylindrical portion **113** of the flow-adjustment piece **111** is press-fitted (inserted) into the medium-diameter hole portion **102B** of the shower cylindrical portion **97**. The flow-adjustment cylindrical portion **113** is press-fitted (inserted) into the medium-diameter hole portion **102B** (or the nozzle hole **102**) of the shower cylindrical portion **97** with a gap between the disk back flat surface **114B** of the flow-adjustment nozzle disk **114** and the second hole step portion **102E** of the nozzle hole **102** in the cylinder center line H of the shower nozzle **3**. At this time, as shown in FIG. **50(a)**, the flow-adjustment piece **11** is press-fitted into the shower cylindrical portion **97** so that one of the flow-adjustment-piece plates **116** is arranged at a center of one of the air introduction passages **112** in the peripheral direction of the shower nozzle **3**.

(334) The flow-adjustment annular plate **115** of the flow-adjustment piece **111** is press-fitted (inserted) into the large-diameter hole portion **102A** of the shower cylindrical portion **97**, and is brought into abutment against the first hole step portion **102D**.

(335) Thus, as shown in FIG. **51**, the flow-adjustment nozzle disk **114** of the flow-adjustment piece **111** is arranged in the air bubble mixing space BR in the shower cylindrical portion **97** at a distance from the disk back flat surface **96B** of the shower nozzle plate **96** in the direction of the cylinder center line H of the shower nozzle **3**. The flow-adjustment nozzle disk **114** and the flow-adjustment annular plate **115** seal the other cylinder end **97B** of the shower cylindrical portion **97** in a liquid-tight manner, and are fixed to the shower cylindrical portion **97**.

(336) As shown in FIG. **50(b)**, the flow-adjustment-piece plates **116** of the flow-adjustment piece **111** are arranged in the air bubble mixing space BR between the shower nozzle plate **96** and the flow-adjustment nozzle disk **114**.

(337) As shown in FIG. **51(b)**, the flow-adjustment-piece plates **116** are arranged to protrude from the flow-adjustment nozzle disk **114** toward the shower nozzle plate **96** in the direction of the cylinder center line H of the shower nozzle **3** (or the cylinder center line J of the flow-adjustment piece **111**) with a mixing gap GP between the disk back flat surface **96B** of the shower nozzle plate **96** and the protruding end **116D**. As shown in FIG. **51(b)**, the flow-adjustment-piece plates **116** are arranged to extend from the plate center line J of the flow-adjustment nozzle disk **114** (or the cylinder center line H of the shower nozzle **3**) toward the shower cylindrical portion **97**. The flow-adjustment-piece plates **116** are arranged with a gap between the inner peripheral surface of the shower cylindrical portion **97** and the flow-adjustment-piece plates **116**.

(338) As shown in FIG. **50(a)**, the liquid throttle holes **117** of the flow-adjustment piece **111** are arranged so that the hole center line M of each of the liquid throttle holes **117** is parallel to the cylinder center line H (the center line) of the shower cylindrical portion **97** (or the shower nozzle **3**). The liquid throttle holes **117** are opened into the air bubble mixing space BR between the shower nozzle plate **96** and the flow-adjustment nozzle disk **114**.

(339) As shown in FIG. **51(b)**, in a space between the protruding end **116D** of each of the flow-adjustment-piece plates **116** and the disk front flat surface **114A** of the flow-adjustment nozzle disk **114** in the direction of the cylinder center line H of the shower nozzle **3**, the air introduction passages **112** are opened into the air bubble mixing space BR in the direction orthogonal to the cylinder center line H of the shower cylindrical portion **97**. As shown in FIG. **50(b)**, the air introduction passages **112** are adjacent to the disk front flat surface **114A** of the flow-adjustment nozzle disk **114**, and are opened into the air bubble mixing space BR.

(340) With this configuration, through the air introduction passages **112**, the air flows into the air

bubble mixing space BR from the direction orthogonal to the hole center line M of each of the liquid throttle holes **117**.

(341) As shown in FIG. **44(b)** and FIG. **51(a)**, each of the air introduction passages **112** is opened into the air bubble mixing space BR as a rectangular hole (an oblong hole) having an opening width (a hole width) AH in the peripheral direction of the shower cylindrical portion **97** (or the shower nozzle **3**) and an opening height (hole height) AL in the direction H of the cylinder center line H of the shower cylindrical portion **97** (or the shower nozzle **3**). The opening width AH of each of the air introduction passages **112** is larger than the plate width HS of each of the flow-adjustment-piece plates **116**.

(342) Thus, as shown in FIG. **50** and FIG. **51**, the air bubble-liquid mixture generating means **4** is arranged so that the flow-adjustment piece **111** is incorporated in the shower nozzle **3** (or in the shower cylindrical portion **97**).

(343) In the shower head X, the mist generating means **5** (the mist generating unit) is configured to form the liquid into the mist of liquid droplets in which the air bubbles are mixed.

(344) As shown in FIG. **1** to FIG. **5**, FIG. **43** to FIG. **45**, and FIG. **52** to FIG. **55**, the mist generating means **5** includes a plurality of mist throttle holes **121**, a mist ring body **122**, and a sealing ring **130**.

(345) As shown in FIG. **42(a)**, FIG. **43**, FIG. **44(b)**, and FIG. **45**, the plurality of mist throttle holes **121** are formed in the shower nozzle plate **96** (or the shower nozzle **3**). The number of the mist throttle holes **121** is, for example, twelve.

(346) As shown in FIG. **43(a)**, the mist throttle holes **12** are arranged in the shower nozzle plate **96** on the outer side of the air bubble-liquid mixture jetting holes **98**. The mist throttle holes **121** are arranged (in a concyclic manner) on a circle CK that has a center along the cylinder center line H (the center line) of the shower nozzle **3** (or the shower cylindrical portion **97**) and is located on the outer side of the air bubble-liquid mixture jetting holes **98**.

(347) As shown in FIG. **43**, the mist throttle holes **121** are arranged at equal angular intervals (equal pitches) of 30 degrees in the peripheral direction of the shower nozzle **3** (or the shower cylindrical portion **97**).

(348) With this configuration, the plurality of mist throttle holes **121** are arranged in the shower nozzle **3** on the outer side of the air bubble-liquid mixture jetting holes **98** (or the air bubble-liquid mixture generating means **4**).

(349) As shown in FIG. **42**, FIG. **43**, FIG. **44(b)**, and FIG. **45**, the mist throttle holes **121** are each formed to pass through the shower nozzle plate **96** in the direction of the cylinder center line H of the shower nozzle **3**, and are opened in the disk front surface **96A** and the disk back flat surface **96B** of the shower nozzle plate **96**. The mist throttle holes **121** are arranged on the outer side of the air introduction passages **112** (or the air bubble-liquid mixture jetting holes **98**) in a direction orthogonal to the cylinder center direction H of the shower nozzle **3**, and are opened into the mist annular space YM.

(350) As shown in FIG. **44(b)**, the mist throttle holes **121** are each formed into a conical hole having a diameter gradually reducing from the disk back flat surface **96B** toward the disk front surface **96A** of the shower nozzle plate **96** in the direction of the cylinder center line H of the shower nozzle **3**.

(351) As shown in FIG. **44**, the mist throttle holes **121** each have a hole length ML in the direction of the cylinder center line H of the shower nozzle **3**. As shown in FIG. **45**, the mist throttle holes **121** each have a hole diameter dM at the disk front surface **96A** of the shower nozzle plate **96**, and has a hole diameter dF at the disk back flat surface **96B** (hole diameter dM>hole diameter dF).

(352) As shown in FIG. **52** to FIG. **55**, the mist ring body **122** includes a guide ring **123** and a plurality of mist guides **124**.

(353) As shown in FIG. **52** to FIG. **55**, the guide ring **123** is made of a synthetic resin and formed into an annular shape. As shown in FIG. **43** and FIG. **54(a)**, the guide ring **123** has a center circle

CL having a ring diameter **D8** equal to the diameter of the circle CK on which the mist throttle holes **121** are arranged.

(354) As shown in FIG. 52 to FIG. 55, the guide ring **123** includes a plurality of guide protrusions **125**. The number of the guide protrusions **125** is, for example, twelve, the same as the number of the mist throttle holes **121**.

(355) The guide protrusions **125** are arranged on the circle CL of the guide ring **123**. The guide protrusions **125** are arranged at equal angular intervals of 30 degrees in the circumferential direction of the guide ring **123**. The guide protrusions **125** are formed to protrude in a direction orthogonal to a center line K of the mist ring body **122** (or the guide ring **123**), and are formed integrally with the guide ring **123**.

(356) As shown in FIG. 52 to FIG. 55, the plurality of mist guides **124** are each made of a synthetic resin and formed into a conical spiral (conical helix or spiral having a truncated cone shape). As shown in FIG. 52(b), the mist guides **124** each include a cone upper surface **124A**, a cone bottom flat surface **124B**, a cone side surface **124C**, and a plurality of spiral surfaces, for example, first and second spiral surfaces **127** and **128** (helical surfaces). The number of the mist guides **124** is twelve, the same as the number of the mist throttle holes **121**.

(357) The first and second spiral surfaces **127** and **128** are each formed into the same spiral shape. The first and second spiral surfaces **127** and **128** are arranged between the cone bottom flat surface **124B** and the cone upper surface **124A** to cross the cone side surface **124C**.

(358) The first and second spiral surfaces **127** and **128** are arranged so as to be point symmetrical with respect to a cone center line L. The second spiral surface **128** is arranged at a position turned about the cone center line L by an angle of 180 degrees from a position of the first spiral surface **127**.

(359) The first and second spiral surfaces **127** and **128** are each formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface **124B** toward the cone upper surface **124A**, and are formed to extend to the cone upper surface **124A**.

(360) The first and second spiral surfaces **127** and **128** are arranged so as to be opposed to each other at the cone upper surface **124A**.

(361) As shown in FIG. 54(a), the mist guides **124** each have a guide height GL in a direction of the cone center line L. The guide height GL is set smaller than the hole length ML of each of the mist throttle holes **121**.

(362) As shown in FIG. 55(a), the mist guides **124** each have a maximum bottom width GH at the cone bottom flat surface **124B**. The maximum bottom width GH is set smaller than the hole diameter dM of each of the mist throttle holes **121**.

(363) As shown in FIG. 52 to FIG. 55, the mist guides **124** are fixed to the guide ring **123**, and are formed integrally with the guide ring **123**. As shown in FIG. 53(a), the mist guides **124** are arranged on the circle CL of the guide ring **123**. Each of the mist guides **124** is arranged so that the cone center line L (guide center line) thereof is located on the circle CL of the guide ring **123**. The mist guides **124** are arranged between the guide protrusions **125** at equal angular intervals of 30 degrees in the peripheral direction of the guide ring **123**. Each of the mist guides **124** is arranged so that a surface end of the first spiral surface **127** and a surface end of the second spiral surface **128** are respectively located at (aligned with) an outer peripheral surface and an inner peripheral surface of the guide ring **123** at the cone bottom flat surface **124B**.

(364) As shown in FIG. 52, FIG. 54(b), and FIG. 55, each of the mist guides **124** is fixed (formed) integrally with the guide ring **123** so that the cone bottom flat surface **124B** is held in abutment against the guide ring **123**. In each of the mist guides **124**, as shown in FIG. 55, the cone bottom flat surface **124B** is fixed to the guide ring **123** so as to protrude from the inner peripheral surface and the outer peripheral surface of the guide ring **123** in the direction orthogonal to the center line K of the mist ring body **122** (or the guide ring **123**).

(365) With this configuration, the mist guides **124** and the guide ring **123** form the mist ring body

122. The mist ring body **122** includes the guide ring **123**, the mist guides **124**, and the guide protrusions **125** formed integrally with each other.

(366) In the mist generating means **5**, as shown in FIG. **56** and FIG. **57**, the mist ring body **122** (including the guide ring **123** and the mist guides **124**) is incorporated in the shower nozzle **3**. As shown in FIG. **56** and FIG. **57**, the mist ring body **122** is arranged concentrically with the shower cylindrical portion **97** with the cylinder center line H (the center line) of the shower nozzle **3** (or the shower cylindrical portion **97**) being a center. The mist ring body **122** is arranged in the mist annular space YM so that the guide ring **123** is externally fitted to the shower cylindrical portion **97**. Thus, the guide ring **123** is arranged on the outer side of the air bubble-liquid mixture jetting holes **98**.

(367) As shown in FIG. **56** and FIG. **57**, the mist ring body **122** is arranged so that the mist guides **124** are inserted in the mist throttle holes **121**, respectively. The mist ring body **122** is arranged so that the cone upper surface **124A** of each of the mist guides **124** is directed toward each of the mist throttle holes **121** in the mist annular space YM.

(368) Each of the mist guides **124** is inserted into each of the mist throttle holes **121** from the cone upper surface **124A**. Each of the mist guides **124** is arranged in each of the mist throttle holes **121** so that the cone center line L is aligned with a hole center line N of each of the mist throttle holes **121**. Each of the mist guides **124** is inserted into each of the mist throttle holes **121** from the cone upper surface **124A** with a gap between the cone side surface **124C** and a conical inner peripheral surface **121A** of each of the mist throttle holes **121**. Each of the mist guides **124** is fitted in each of the mist throttle holes **121** so that the cone bottom flat surface **124B** side (or the cone side surface **124C** on the cone bottom flat surface **124B** side) is held in abutment against the conical inner peripheral surface **121A** of each of the mist throttle holes **121**.

(369) Thus, each of the mist guides **124** is fitted in each of the mist throttle holes **121** so as to define first and second mist flow passages **51** and **52** each having a spiral shape between the first and second spiral surfaces **127** and **128** and the conical inner peripheral surface **121A** of each of the mist throttle holes **121** and between the cone side surface **124C** and the conical inner peripheral surface **121A**. Each of the mist guides **124** and each of the mist throttle holes **121** define the first and second mist flow passages **51** and **52** each having a spiral shape (helical shape) along the first and second spiral surfaces **127** and **128**. As shown in FIG. **57(b)**, the first and second mist flow passages **51** and **52** are each defined in a spiral shape between the first and second spiral surfaces **127** and **128** and the conical inner peripheral surface **121A** of each of the mist throttle holes **121** and between the cone side surface **124C** of each of the mist guides **124** and the conical inner peripheral surface **121A**. The first and second mist flow passages **51** and **52** are each defined in a spiral shape to extend from the cone bottom flat surface **124B** to the cone upper surface **124A** of the mist guide **124** in the direction of the cylinder center line H of the shower nozzle **3**, and are opened in each of the mist throttle holes **121** and in the disk back flat surface **96B** of the shower nozzle plate **96**.

(370) As shown in FIG. **56** and FIG. **57**, along with insertion of the mist guides **124** into the mist throttle holes **121**, the guide ring **123** and the guide protrusions **125** are brought into abutment against the disk back flat surface **126B** of the shower nozzle plate **96** through the mist annular space YM.

(371) As shown in FIG. **57(a)**, the sealing ring **130** is externally fitted to the shower cylindrical portion **97** of the shower nozzle **3**, and is brought into abutment against the sealing step portion **101**. The sealing ring **130** is externally fitted to the shower cylindrical portion **97** so as to protrude from the outer peripheral surface of the shower cylindrical portion **97** into the mist annular space YM in the direction orthogonal to the cylinder center line H of the shower nozzle **3**.

(372) Thus, the sealing ring **130** is freely brought into abutment against the guide protrusions **125** of the mist ring body **122**, thereby preventing the mist ring body **122** from slipping off.

(373) As shown in FIG. **50**, FIG. **51**, FIG. **56**, and FIG. **57**, the flow-adjustment piece **111** and the

mist ring body **122** (including the guide ring **123** and the mist guides **124**) are incorporated in the shower nozzle **3**. Thus, the shower nozzle **3**, the air bubble-liquid mixture generating means **4**, and the mist generating means **5** form a nozzle unit NU.

(374) As shown in FIG. **58** to FIG. **60**, the nozzle unit NU (including the shower nozzle **3**, the air bubble-liquid mixture generating means **4**, and the mist generating means **5**) is arranged in the flow passage switching means **2** (or in the switching handle **21**) fitted in the shower main body **1** (or the head portion **7**).

(375) As shown in FIG. **58**, the nozzle unit NU is arranged so that the flow-adjustment piece **111** (or the disk back flat surface **114B** of the flow-adjustment nozzle disk **114**) is directed toward the large-diameter hole portion **33A** (or the handle hole **33**) of the switching handle **21**. The nozzle unit NU is arranged concentrically with the switching handle **21** with the cylinder center line B of the switching handle **21** being a center.

(376) As shown in FIG. **58**, the nozzle unit NU is inserted into the large-diameter hole portion **33A** of the switching handle **21** from the other cylinder end **95B** of the nozzle outer cylindrical portion **95** of the shower nozzle **3**.

(377) The nozzle unit NU is arranged so that the threaded portion **100** of the shower nozzle **3** is screwed into the threaded portion **34** of the switching handle **21**. Through turning of the nozzle unit NU, the nozzle outer cylindrical portion **95** of the shower nozzle **3** is received in the large-diameter hole portion **33A** (or in the handle hole) of the switching handle **21**. The shower nozzle **3** is turned until the other cylinder end **95B** of the nozzle outer cylindrical portion **95** is brought into abutment against the first valve element protruding portions **80** of the switching valve element **27**.

(378) At this time, the sealing ring **103** of the shower nozzle **3** is brought into in press-contact with the large-diameter hole portion **33A** of the switching handle **21**, thereby sealing the large-diameter hole portion **33A** in a liquid-tight manner.

(379) Thus, the shower nozzle **3** of the nozzle unit NU is fixed to the switching handle **21**, and is fitted to the other end **1B** of the shower main body **1**.

(380) In the shower nozzle **3**, the shower nozzle plate **96** defines a liquid inflow space RP in the outflow passage **10**. The liquid inflow space RP is a space sealed in a liquid-tight manner, and the liquid is caused to flow into the liquid inflow space RP through the outflow passage **10**.

(381) In the nozzle unit NU, as shown in FIG. **58**, the shower cylindrical portion **97** of the shower nozzle **3** and the flow-adjustment piece **111** are inserted into the large-diameter hole portion **87A** (or into the shower outflow hole **87**/into the second valve element cylindrical portion **73**) of the switching valve element **27** within the liquid inflow space RP. The shower cylindrical portion **97** and the flow-adjustment piece **111** are arranged with a gap between the other cylinder end **97B** and the valve element disk **74** (or the disk front flat surface **74A**) in the direction of the cylinder center line F of the switching valve element **27**. Within the liquid inflow space RP, the sealing ring **130** of the shower nozzle **3** is inserted into the large-diameter hole portion **87A** (or into the shower outflow hole **87**) of the switching valve element **27**, and is brought into abutment against the hole step portion **87C** of the switching valve element **27**. In the large-diameter hole portion **87A**, the sealing ring **130** is brought into press-contact with the inner peripheral surface of the second valve element cylindrical portion **73**, thereby sealing the large-diameter hole portion **87A** of the switching valve element **27** in a liquid-tight manner.

(382) Thus, the shower cylindrical portion **97** of the shower nozzle **3** is inserted in the large-diameter hole portion **87A** (or the shower outflow hole **87**) of the switching valve element **27** so as to protrude toward the outflow passage **10** side (into the liquid inflow space RP). The liquid (liquid in a liquid inflow space PR) having flowed out through the outflow passage **10** and having flowed out through the switching valve element **27** is caused to flow from the other cylinder end **97B** (or the liquid throttle holes **117** of the flow-adjustment piece **111**) into the air bubble mixing space BR in the shower cylindrical portion **97**.

(383) When the shower nozzle **3** of the nozzle unit NU is fixed to the switching handle **21**, the

shower nozzle **3**, the flow-adjustment piece **111** (of the air bubble-liquid mixture generating means **4**), the mist ring body **122** (of the mist generating means **5**), and the switching valve element **27** are freely turnable together with the switching handle **21** with respect to the switching valve seat element **25**, the switching base **22**, and the shower main body **1**.

(384) As shown in FIG. **58**, the flow-adjustment piece **111** of the air bubble-liquid mixture generating means **4** is arranged with a gap between the valve element disk **74** (or the disk front flat surface **74A**) of the switching valve element **27** and the flow-adjustment piece **111**, and is inserted in the large-diameter hole portion **87A** (or in the second valve element cylindrical portion **73**) of the switching valve element **27**.

(385) Thus, as shown in FIG. **60**, the liquid throttle holes **117** are each opened to the outflow passage **10** side (in the liquid inflow space RP), and are each opened in the large-diameter hole portion **87A** of the switching valve element **27** and the air bubble mixing space BR. The liquid (liquid in the liquid inflow space RP) having flowed out through the outflow passage **10** and having flowed out through the switching valve element **27** is jetted into the air bubble mixing space BR through the liquid throttle holes **117**.

(386) As shown in FIG. **58**, the flow passage switching means **2** is arranged between the flow-adjustment piece **111** of the air bubble-liquid mixture generating means **4** and the outflow passage **10** and in the outflow passage **10** of the shower main body **1**.

(387) In the flow passage switching means **2**, the switching valve seat element **25** and the switching valve element **27** are arranged between the flow-adjustment piece **111** and the outflow passage **10** and within the liquid inflow space RP, and the switching base **22** is arranged in the outflow passage **10**.

(388) As shown in FIG. **59**, the mist generating means **5** is configured to form the liquid (liquid caused to flow out through the outflow passage **10**) caused to flow into the mist generating means **5** through the flow passage switching means **2** (or the switching valve element **27**) into the mist of liquid droplets in which the air bubbles are mixed.

(389) In the mist generating means **5**, the mist throttle holes **121** are each opened to the outflow passage **10** side and in the liquid inflow space RP between the shower nozzle plate **96** and the flow passage switching means **2** (or the switching valve element **27**).

(390) With this configuration, the mist throttle holes **121** are each formed to pass through the shower nozzle plate **96** while gradually reducing a diameter from the outflow passage **10** side (or the liquid inflow space BR side).

(391) The mist throttle holes **121** each communicate with the outflow passage **10** through the outer outflow holes **82** of the switching valve element **27**, the valve seat holes **64** and **65** of the switching valve seat element **25**, and the base inflow passages Z of the switching base **22** (or the liquid inflow space PR).

(392) In the mist generating means **5**, as shown in FIG. **59**, the mist ring body **122** is arranged so that the guide ring **123** is held in abutment against the one cylinder end **73A** of the second valve element cylindrical portion **73**.

(393) The guide ring **123** and the guide protrusions **125** are brought into abutment against the disk back flat surface **96B** of the shower nozzle plate **96** from the outflow passage **10** side (or the liquid inflow space PR side or the mist annular space YM side).

(394) As shown in FIG. **59**, the first and second mist flow passages **51** and **52** are opened between the flow passage switching means **2** and the shower nozzle **3**, and communicate with the outflow passage **10**.

(395) When the shower nozzle **3** is turned, the switching valve element **27** and the switching valve seat element **25** are pressed toward the switching base **22** side, thereby compressing the coil spring **30**. The compressed coil spring **30** urges the switching valve seat element **25** to the switching valve element **27** by a spring force, thereby bringing the valve seat disk **63** (or the disk front flat surface **63A**) into press-contact with the sealing rings **28** of the cylindrical valve elements **76** and **77**.

(396) Thus, the sealing rings **28** connect the valve element hole **88** of the cylindrical valve element **76** and the valve element hole **90** of the cylindrical valve element **77** to the valve seat holes **64** and **65** in a liquid-tight manner, respectively.

(397) When the nozzle unit NU (including the shower nozzle **3**, the air bubble-liquid mixture generating means **4**, and the mist generating means **5**) and the flow passage switching means **2** (including the switching handle **21**, the switching base **22**, the switching valve seat element **25**, and the switching valve element **27**) are thus mounted to the shower main body **1** (or the head portion **7**), as shown in FIG. **1** to FIG. **3** and FIG. **58** to FIG. **60**, the shower head X is at the shower position **P1**.

(398) At the shower position **P1**, as shown in FIG. **1** to FIG. **3** and FIG. **58** to FIG. **60**, the switching handle **21** is arranged so that the shower protruding portion **38** overlaps the reference protruding portion **14** (or the highest point **7a**) of the shower main body **1**.

(399) At the shower position **P1**, as shown in FIG. **40**, the switching valve element **27** is arranged so that the valve element hole **88** of the cylindrical valve element **76** and the valve element hole **90** of the cylindrical valve element **77** are opened (valve-opened) toward the valve seat holes **64** and **65** of the switching valve seat element **25**, respectively.

(400) At the shower position **P1**, the flow passage switching means **2** connects the liquid throttle holes **117** of the air bubble-liquid mixture generating means **4** to the outflow passage **10**. The liquid throttle holes **117** of the flow-adjustment piece **111** each communicate with the outflow passage **10** of the shower main body **1** through the valve element flow passages **78** and **79** and the valve element holes **88** and **90** of the switching valve element **27**, the valve seat holes **64** and **65** of the switching valve seat element **25**, and the base inflow passages **Z** of the switching base **22**.

(401) At the shower position **P1**, as shown in FIG. **41**, the switching valve element **27** is arranged so that the valve element regulating flat surface **83A** of one of the first handle regulating protruding portions **83** and the valve element regulating flat surface **85A** of one of the second handle regulating protruding portions **85** are held in abutment against the first base regulating flat surface **59A** of the base protrusion **59** and the fourth base regulating flat surface **60B** of the base protrusion **60** of the switching base **22**, respectively.

(402) As shown in FIG. **2**, FIG. **58**, and FIG. **59**, the shower head X at the shower position **P1** causes the liquid to flow into the inflow passage **9** of the shower main body **1** (or the handle portion **6**).

(403) The liquid having flowed into the inflow passage **9** is caused to flow into the outflow passage **10**. Through the outflow passage **10**, the liquid having flowed from the inflow passage **9** is caused to flow out. As shown in FIG. **37** and FIG. **59**, the liquid flows through the outflow passage **10** into the base inflow passages **Z** of the switching base **22**, and then flows into the valve seat holes **64** and **65** of the switching valve seat element **25** in the liquid inflow space **PR**.

(404) As shown in FIG. **59**, the liquid having flowed into the valve seat holes **64** and **65** flows into the valve element hole **88** of the cylindrical valve element **76** and the valve element hole **89** of the cylindrical valve element **77** of the switching valve element **27**. In the switching valve element **27**, as shown in FIG. **39**, the liquid flows from the valve element holes **88** and **89** through the valve element flow passages **78** and **79** each having a helical shape into the shower outflow hole **87** in the second valve element cylindrical portion **73**.

(405) At this time, as shown in FIG. **39**, the liquid is caused to helically flow through the valve element flow passages **78** and **79** each having a helical shape, and flow into the entire shower outflow hole **87** of the second valve element cylindrical portion **73**.

(406) The liquid having flowed into the shower outflow hole **87** is jetted into the air bubble mixing space **BR** through the liquid throttle holes **117** of the flow-adjustment piece **111** (or the air bubble-liquid mixture generating means **4**). Thus, through the liquid throttle holes **117**, the liquid having flowed out through the outflow passage **10** is jetted into the air bubble mixing space **BR**.

(407) At this time, as shown in FIG. **60**, through the liquid throttle holes **117** of the flow-adjustment

piece **111**, the liquid in the shower outflow hole **87** (or in the liquid inflow space **PR**) is jetted into the air bubble mixing space **BR** toward the air bubble-liquid mixture jetting holes **98** of the shower nozzle plate **96**. The liquid is jetted between the flow-adjustment-piece plates **116** in the air bubble mixing space **BR**. In the air bubble mixing space **BR**, the liquid is jetted between the shower nozzle plate **96** and the flow-adjustment nozzle disk **114** while flowing (being adjusted in flow) in parallel to the cylinder center line **H** of the shower cylindrical portion **97** (or the shower nozzle **3**).

(408) When the liquid is jetted into the air bubble mixing space **BR**, due to the jet of the liquid, the air is introduced through the air introduction passages **112** into the air bubble mixing space **BR**. Through the air introduction passages **112**, the air is caused to flow between the flow-adjustment-piece plates **116** in the air bubble mixing space **BR**.

(409) As shown in FIG. **60**, in the air bubble mixing space **BR**, the air introduction passages **112** cause the air to flow toward the disk front flat surface **74A** of the valve element disk **74** that is adjacent to the liquid throttle holes **117** of the flow-adjustment piece **111**. In the air bubble mixing space **BR**, the air is caused to flow (jet) between the flow-adjustment-piece plates **116** of the flow-adjustment piece **111** through the air introduction passages **112**. The air is caused to flow (jet) into the air bubble mixing space **BR** from the direction orthogonal to the hole center line **M** of each of the liquid throttle holes **117**.

(410) Thus, the air introduced into the air bubble mixing space **BR** is mixed into the liquid at the same time as the liquid is jetted through the liquid throttle holes **117**.

(411) In the air bubble mixing space **BR**, the liquid and the air flow turbulently by being introduced by the protruding ends **116D** along the flow inclined surfaces **118** of the flow-adjustment-piece plates **116**, and then flow into the mixing gap **GP** between the protruding ends **116D** of the flow-adjustment-piece plates **116** and the shower nozzle plate **96**.

(412) Thus, on the protruding end **116D** side protruding toward the shower nozzle **3** (or the shower nozzle plate **96**), each of the flow-adjustment-piece plates **116** causes the liquid jetted through the liquid throttle holes **117** to flow turbulently and flow into the mixing gap **GP**.

(413) In the mixing gap **GP** within the air bubble mixing space **BR**, due to the turbulent flow, the air mixed into the liquid is broken (divided) into micrometer-sized air bubbles (microbubbles) and nanometer-sized air bubbles (ultrafine bubbles).

(414) The micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) mix with and dissolve in the liquid.

(415) The liquid (air bubble-liquid mixture), in which the micrometer-sized air bubbles and the nanometer-sized air bubbles are mixed, is jetted to the outside through the air bubble-liquid mixture jetting holes **98** of the shower nozzle plate **96**. Through the air bubble-liquid mixture jetting holes **98**, the air bubble-liquid mixture is jetted out of the air bubble mixing space **BR**.

(416) As shown in FIG. **61**, in the shower head **X** at the shower position **P1**, the switching handle **21** is turned by an angle of 90 degrees with respect to the shower main body **1** (or the switching base **22** and the switching valve seat element **25**) so that the mist protruding portion **39** is arranged at the reference protruding portion **14** of the shower main body **1**.

(417) The switching valve element **27** (of the flow passage switching means **2**), the shower nozzle **3**, the flow-adjustment piece **111** (of the air bubble-liquid mixture generating means **4**), and the mist ring body **122** (of the mist generating means **5**) are turned at the same time as the switching handle **21** is turned.

(418) Thus, the shower head **X** is turned from the shower position **P1** to the mist position **P2**.

(419) At the mist position **P2**, as shown in FIG. **63** and FIG. **64**, the valve element hole **88** of the cylindrical valve element **76** and the valve element hole **90** of the cylindrical valve element **77** of the switching valve element **27** are closed (valve-closed) by the valve seat disk **63** (or the disk front flat surface **63A**) of the switching valve seat element **25**.

(420) At this time, along with turning of the switching valve element **27**, the sealing rings **28** are brought into slide-contact with the valve seat disk **63** (or the disk front flat surface **63A**) of the

switching valve seat element **25** so that the cylindrical valve elements **76** and **77** are closed. Due to the spring force of the coil spring **30**, the valve seat disk **63** of the switching valve seat element **25** is held in press-contact with the sealing rings **28** in the closed cylindrical valve elements **76** and **77**. (421) Thus, the sealing rings **28** seal the valve element holes **88** and **90** in a liquid-tight manner, and block (close) the valve element holes **88** and **90** from the valve seat holes **64** and **65** of the switching valve seat element **25**.

(422) At the mist position **P2**, the flow passage switching means **2** connects the mist throttle holes **121** (of the mist ring body **122**) of the mist generating means **5** to the outflow passage **10**. The mist throttle holes **121** (of the mist ring body **122**) communicate with the outflow passage **10** of the shower main body **1** through the liquid inflow space **RP** between the switching valve element **27** and the shower nozzle **3**, the outer outflow holes **82** of the switching valve element **27**, the valve seat holes **64** and **65** of the switching valve seat element **25**, and the base inflow passages **Z** of the switching base **22**.

(423) At the mist position **P2**, as shown in FIG. **65**, the switching valve element **27** is arranged so that the valve element regulating flat surface **83A** of one of the first handle regulating protruding portions **83** and the valve element regulating flat surface **85A** of one of the second handle regulating protruding portions **85** are held in abutment against the second base regulating flat surface **59B** of the base protrusion **59** and the third base regulating flat surface **60A** of the base protrusion **60** of the switching base **22**, respectively.

(424) As shown in FIG. **62**, the shower head **X** at the mist position **P2** causes the liquid to flow into the inflow passage **9** of the shower main body **1** (or the handle portion **6**).

(425) The liquid having flowed into the inflow passage **9** is caused to flow into the outflow passage **10**. Through the outflow passage **10**, the liquid having flowed from the inflow passage **9** is caused to flow out. As shown in FIG. **37** and FIG. **62**, the liquid flows through the outflow passage **10** into the base inflow passages **Z** of the switching base **22**, and then flows into the valve seat holes **64** and **65** of the switching valve seat element **25** in the liquid inflow space **PR**.

(426) As shown in FIG. **62**, the liquid having flowed into the valve seat holes **64** and **65** flows through the outer outflow holes **82** of the switching valve element **27** into the liquid inflow space **PR** in the shower nozzle plate **96**.

(427) The liquid flows through the liquid inflow space **PR** into the mist throttle holes **121**.

(428) As shown in FIG. **66**, the liquid having flowed into the mist throttle holes **121** flows through the first and second mist flow passages **51** and **52** each having a spiral shape, and then flows out of the mist throttle holes **121**. Further, the mist of liquid droplets is jetted to the outside through the mist throttle holes **121**.

(429) The liquid is increased in pressure by flowing through the first and second mist flow passages **51** and **52** each having a spiral shape, and is jetted into the mist throttle holes **121** through the first and second mist flow passages **51** and **52**.

(430) Thus, the liquid jetted into the mist throttle holes **121** through the first and second mist flow passages **51** and **52** flows turbulently at high pressure. Further, when the mist of liquid droplets is jetted through the mist throttle holes **121**, an outlet side of each of the mist throttle holes **121** (side from which the mist of liquid droplets is jetted) is brought into a negative pressure state.

(431) With the outlet side of each of the mist throttle holes **121** brought into the negative pressure state, when the liquid, which is jetted into the mist throttle holes **121** through the first and second mist flow passages **51** and **52** and flows turbulently at high pressure, passes through the outlet portion of each of the mist throttle holes **121**, the air bubbles are separated due to reduced pressure, and the air that is taken in at the time of jetting is broken (divided) by the turbulent flow. Thus, the liquid is formed into the mist of liquid droplets in which the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) are mixed and dissolved.

(432) Further, at the cone upper surface **124A** of each of the mist guides **124**, the liquid is jetted into each of the mist throttle holes **121** through the first and second mist flow passages **51** and **52**

opposed to each other, and collides with the mist guide **124** and the shower nozzle plate **96**, thereby being formed into the mist of liquid droplets in which a sufficient volume of air bubbles is mixed. The mist of liquid droplets in which the air bubbles are mixed is jetted through each of the mist throttle holes **121**. Through each of the mist throttle holes **121**, the mist of liquid droplets in which the air bubbles are mixed is jetted to the outside.

(433) Thus, the mist generating means **5** forms the liquid having flowed out through the outflow passage **10** into the mist of liquid droplets in which the air bubbles are mixed.

(434) The shower head X is thus set to the shower position **P1** or the mist position **P2** through forward and reverse turning of the switching handle **21** within an angle range of 90 degrees.

(435) At this time, as shown in FIG. **41** and FIG. **65**, the base protrusions **59** and **60** of the switching base **22** and the first and second handle regulating protruding portions **83** and **85** of the switching valve element **27** regulate the turning of the switching handle **21** within the angle range of 90 degrees.

(436) When the shower head X is switched to the shower position **P1** or the mist position **P2**, the shower head X can jet the air bubble-liquid mixture at the shower position **P1**, and can jet the mist of liquid droplets, in which the air bubbles are mixed, at the mist position **P2**.

(437) In the shower head X, the number of the flow-adjustment-piece plates **116** is not limited to four. It is only required that the number of the flow-adjustment-piece plates **116** be plural, for example, three, five, or six and so on. The plurality of flow-adjustment-piece plates **116** are formed on the flow-adjustment nozzle disk **114** at equal intervals in the peripheral direction of the flow-adjustment nozzle disk **114**.

(438) In the shower head X, the number of the spiral surfaces of the mist guide **124** is not limited to two. It is only required that the number of the spiral surfaces of the mist guide **124** be plural, for example, three, four, or five and so on. The plurality of spiral surfaces are formed on the mist guide **124** (or the cone side surface **124C**) at equal intervals in the peripheral direction with the cone center line L of the mist guide **124** being a center.

EXAMPLES

(439) For the shower head X, a “shower test” of generating the air bubble-liquid mixture (air bubble-water mixture) was carried out under a condition in which the shower nozzle **3** and the liquid generating means **4** (including the flow-adjustment piece **111** and the air introduction passages **112**) were used.

(440) For the shower head X, a “mist test” of generating the mist of liquid droplets (mist of water droplets) was carried out under a condition in which the mist generating means **5** (including the mist throttle holes **121** and the mist guides **124**) was used.

(441) In the “shower test” and the “mist test”, similarly to the description with reference to FIG. **26** to FIG. **41**, the flow passage switching means **2** (including the switching handle **21**, the switching base **22**, the switching valve seat element **25**, and the switching valve element **27**) was arranged in the shower main body **1**.

(442) <1>“Shower Test”

(443) The “shower test” was carried out in Example 1, Example 2, Example 3, and Comparative Example 1.

(444) (1) Shower Nozzle

(445) The “shower nozzle **3**” was common (the same) in Example 1, Example 2, Example 3, and Comparative Example 1.

(446) The “shower nozzle **3**” in Example 1, Example 2, Example 3, and Comparative Example 1 is described with reference to FIG. **43** to FIG. **45**.

(447) The “shower nozzle **3**” in Example 1, Example 2, Example 3, and Comparative Example 1 has the following configuration. Total number of the air bubble-liquid mixture jetting holes **98**: 36
Hole diameter of each of the air bubble-liquid mixture jetting holes **98** (conical hole): 1.4 mm (opened in the disk front surface **96A**)
Hole diameter of each of the air bubble-liquid mixture

jetting holes **98** (conical hole): 1.8 mm (opened in the disk back flat surface **96B**.) Radius **r3** of the circle **CD**: 3.5 mm. Radius **r4** of the circle **CE**: 6.2 mm. Radius **r5** of the circle **CF**: 8.7 mm. Number of the air bubble-liquid mixture jetting holes **98** arranged on the circle **CD**: 6 (arranged at equal pitches in the peripheral direction of the shower cylindrical portion **97**) Number of the air bubble-liquid mixture jetting holes **98** arranged on the circle **CE**: 12 (arranged at equal pitches in the peripheral direction of the shower cylindrical portion **97**) Number of the air bubble-liquid mixture jetting holes **98** arranged on the circle **CF**: 18 (arranged at equal pitches in the peripheral direction of the shower cylindrical portion **97**) Inner diameter **d5** of the small-diameter hole portion of the handle hole **33**: 6.2 mm

(2) Flow-adjustment Piece

(448) The “flow-adjustment piece **111**” in Example 1 is described with reference to FIG. **47**, FIG. **48**, and FIG. **67**.

(449) The “flow-adjustment piece **111**” in Example 1 has the following configuration. Total number of the liquid throttle holes **117**: 40 Hole diameter **da** of each of the liquid throttle holes **117**: 0.6 mm (opened in the disk front flat surface **114A**) Hole diameter **db** of each of the liquid throttle holes **117**: 1.0 mm (opened in the disk back flat surface **114B**) Radius **r6** of the circle **CG**: 4.0 mm Radius **r7** of the circle **CH**: 6.0 mm Radius **r8** of the circle **CI**: 9.0 mm Number of the liquid throttle holes **117** arranged on the circle **CG**: 8 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that two holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle **CH**: 12 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that three holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle **CI**: 20 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that five holes are formed in each region between the flow-adjustment-piece plates **116**) Piece height of the flow-adjustment piece **111**: 8.2 mm Number of the flow-adjustment-piece plates **116**: 4 (arranged at equal angular intervals of 90 degrees in the peripheral direction of the flow-adjustment nozzle disk **114**) Plate width **HS** of the flow-adjustment-piece plate **116**: 4.0 mm Plate length **LS** of the flow-adjustment-piece plate **116**: 9.2 mm Plate thickness **TS** of the flow-adjustment-piece plate **116**: 1.4 mm Radius **rX** (of the arc) of the flow inclined surface **118**: 1.0 mm

(450) The “flow-adjustment piece **111**” in Example 2 is described with reference to FIG. **47**, FIG. **48**, and FIG. **68**.

(451) The “flow-adjustment piece **111**” in Example 2 has the following configuration. Total number of the liquid throttle holes **117**: 48

(452) Hole diameter **da** of each of the liquid throttle holes **117**: 0.6 mm (opened in the disk front flat surface **114A**) Hole diameter **db** of each of the liquid throttle holes **117**: 1.0 mm (opened in the disk back flat surface **114B**) Radius **r6** of the circle **CG**: 2.0 mm Radius **r7** of the circle **CH**: 4.0 mm Radius **r8** of the circle **CI**: 6.0 mm Radius **r9** of the circle **CM**: 9.0 mm Number of the liquid throttle holes **117** arranged on the circle **CG**: 4 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that one hole is formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle **CH**: 8 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that two holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle **CI**: 16 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that four holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle **CM**: 20 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that five holes are formed in each region between the flow-adjustment-piece plates **116**)

(453) The “flow-adjustment piece **111**” in Example 2 is the same as the “flow-adjustment piece

111” in Example 1 with regard to the piece height of the flow-adjustment piece **111**, the number of the flow-adjustment-piece plates **116**, the plate width HS of each of the flow-adjustment-piece plates **116**, the plate length LS of each of the flow-adjustment-piece plates **116**, the plate thickness TS of each of the flow-adjustment-piece plates **116**, and the radius rX (of the arc) of the flow inclined surface **118**.

(454) The “flow-adjustment piece **111**” in Example 3 is described with reference to FIG. 47, FIG. 48, and FIG. 69.

(455) The “flow-adjustment piece **111**” in Example 3 has the following configuration. Total number of the liquid throttle holes **117**: 52 Hole diameter da of each of the liquid throttle holes **117**: 0.6 mm (opened in the disk front flat surface **114A**) Hole diameter db of each of the liquid throttle holes **117**: 1.0 mm (opened in the disk back flat surface **114B**) Radius r6 of the circle CG: 2.0 mm Radius r7 of the circle CH: 4.0 mm Radius r8 of the circle CI: 6.0 mm Radius r9 of the circle CM: 9.0 mm Number of the liquid throttle holes **117** arranged on the circle CG: 4 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that one hole is formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle CH: 8 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that two holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle CI: 16 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that four holes are formed in each region between the flow-adjustment-piece plates **116**) Number of the liquid throttle holes **117** arranged on the circle CM: 24 (arranged at equal pitches in the peripheral direction of the flow-adjustment nozzle disk **114** so that six holes are formed in each region between the flow-adjustment-piece plates **116**)

(456) The “flow-adjustment piece **111**” in Example 3 is the same as the “flow-adjustment piece **111**” in Example 1 with regard to the piece height of the flow-adjustment piece **111**, the number of the flow-adjustment-piece plates **116**, the plate width HS of each of the flow-adjustment-piece plates **116**, the plate length LS of each of the flow-adjustment-piece plates **116**, the plate thickness TS of each of the flow-adjustment-piece plates **116**, and the radius rX (of the arc) of the flow inclined surface **118**.

(457) Unlike the “flow-adjustment piece” in Example 1, Example 1, and Example 3, the “flow-adjustment piece” in Comparative Example 1 is a “flow-adjustment piece without a flow-adjustment-piece plate”, in which the flow-adjustment-piece plates are not formed on the flow-adjustment nozzle disk.

(458) The “flow-adjustment piece” in Comparative Example 1 is the same as that in Example 1 with regard to the number of the liquid throttle holes, the hole diameter of each of the liquid throttle holes, the radii r6 to r8 of the circles CG to CI, and the number of the liquid throttle holes arranged on each of the circles CG to CI.

(459) (3) Air Introduction Passage

(460) The “air introduction passage **112**” is common (the same) in Example 1, Example 2, Example 3, and Comparative Example 1.

(461) The “air introduction passage **112**” in Example 1, Example 2, Example 3, and Comparative Example 1 is described with reference to FIG. 43 and FIG. 44.

(462) The “air introduction passage **112**” in Example 1, Example 2, Example 3, and Comparative Example 1 has the following configuration. Number of the air introduction passages: 3 Radius of the circle CJ: 12.25 mm

(463) The air introduction passages **112** were arranged on the circle CJ, and were arranged at equal angular intervals (equal pitches) of 120 degrees in the circumferential direction of the circle CJ (or the shower nozzle 3).

(464) (4) Air Bubble Mixing Space and Mixing Gap

(465) Similarly to the description with reference to FIG. 50 and FIG. 51, the “flow-adjustment

piece” in Example 1, Example 2, Example 3, and Comparative Example 1 was inserted in the air bubble mixing space BR (or in the shower cylindrical portion 97), and was fixed to the shower nozzle 3.

(466) The “air bubble mixing space BR” is common (the same) in Example 1, Example 2, Example 3, and Comparative Example 1, and has the following configuration. Hole diameter d5 of the air bubble mixing space: 6.2 mm Hole length LK of the air bubble mixing space: 7.0 mm

(467) The “mixing gap GP” is common (the same) in Example 1, Example 2, and Example 3, and has the following configuration. Mixing gap GP: 2.8 mm

(5) Arrangement and Opening Dimension of Air Introduction Passage

(468) Similarly to the description with reference to FIG. 44 and FIG. 51, the “air introduction passage” in Example 1, Example 2, Example 3, and Comparative Example 1 was opened adjacent to the flow-adjustment nozzle disk 114 (or the disk front flat surface 114A).

(469) The “air introduction passage” in Example 1, Example 2, Example 3, and Comparative Example 1 has the following configuration. Opening width AH: 5.05 mm Opening height AL: 0.8 mm

(470) The opening width is a dimension in the peripheral direction of the shower cylindrical portion. The opening height is a dimension in the direction of the cylinder center line of the shower cylindrical portion.

(471) (6) Liquid, Static Liquid Pressure (Hydrostatic Pressure) of Liquid, and Liquid Feeding Rate (Water Feeding Rate)

(472) The “liquid”, “static liquid pressure (hydrostatic pressure) of the liquid”, and a “liquid feeding rate (water feeding rate)” are the same in Example 1, Example 2, Example 3, and Comparative Example 1.

(473) In Example 1, Example 2, Example 3, and Comparative Example 1, the following is employed. Liquid: tap water (water), Static liquid pressure (hydrostatic pressure) of the liquid (water): 0.2 MPa (megapascals) Liquid feeding rate (water feeding rate) of the liquid (water): 9.2 liters/minute (9.2 liters per minute)

(474) In Example 1, Example 2, Example 3, and Comparative Example 1, under a condition in which the “hydrostatic pressure” was 0.2 MPa and the “water feeding rate” was 9.2 liters/minute, the tap water was caused to flow into the inflow passage and jetted through the air bubble-liquid mixture jetting holes.

(475) (7) Measurement of Quantity of Air Bubbles

(476) In the “shower test”, the air bubble-water mixture was jetted through the air bubble-liquid mixture jetting holes, and a quantity of air bubbles mixed into the air bubble-water mixture was measured.

(477) In Example 1, the quantity of air bubbles (bubble quantity) including the micrometer-sized air bubbles (microbubbles) and the nanometer-sized air bubbles (ultrafine bubbles) was measured in a case in which the air bubble-water mixture was jetted at a rate of 8 liters/minute and a rate of 10 liters/minute.

(478) In Example 2, the quantity of air bubbles (bubble quantity) including the microbubbles and the ultrafine bubbles was measured in a case in which the air bubble-water mixture was jetted at a rate of 10 liters/minute.

(479) In Example 3, the quantity of air bubbles (bubble quantity) including the microbubbles and the ultrafine bubbles was measured in a case in which the air bubble-water mixture was jetted at a rate of 10 liters/minute.

(480) In Comparative Example 1, the quantity of air bubbles (bubble quantity) including the microbubbles and the ultrafine bubbles was measured in a case in which the air bubble-water mixture was jetted at a rate of 10 liters/minute.

(481) In Example 1, Example 2, Example 3, and Comparative Example 1, the quantity of air bubbles (bubble quantity) contained per a milliliter (ml) of the air bubble-water mixture was

measured.

(482) In Example 1, Example 2, Example 3, and Comparative Example 1, a total quantity of microbubbles and a microbubble diameter of the microbubbles largest in quantity were measured.

(483) In Example 1, Example 2, Example 3, and Comparative Example 1, a total quantity of ultrafine bubbles and an ultrafine bubble diameter of ultrafine bubbles largest in quantity were measured.

(484) In Example 1, a minimum microbubble diameter and a quantity of microbubbles each having the minimum microbubble diameter were measured.

(485) Measurement results of the microbubbles in Example 1, Example 2, Example 3, and Comparative Example 1 are shown in “Table 1”.

(486) TABLE-US-00001 TABLE 1 Measurement results of microbubbles in “shower test”
Diameter of the Quantity of the microbubbles microbubbles largest in largest in Total quantity
quantity quantity of microbubbles Example 1 28.67 6,060 8,492 (10 L/min.) Example 1 29.12
3,918 4,634 (8 L/min.) Example 2 27.75 2,653 3,509 Example 3 27.92 4,707 6,023 Comparative
7.19 595 1,722 Example 1 Diameter of the microbubbles: micrometer Quantity of the microbubbles
largest in quantity: number of the microbubbles/milliliter Total quantity of microbubbles: total
number of microbubbles/milliliter

(487) In Example 1, the minimum microbubble diameter was 4.44 micrometers (μm), and a quantity of the microbubbles smallest in quantity was 1,200/milliliter.

(488) In Example 1, as shown in “Table 1”, at the rate of 10 liters/minute, the diameter of the microbubbles largest in quantity was 28.67 micrometers (μm), the quantity of the microbubbles largest in quantity was 6,060/milliliter, and the total quantity of microbubbles was 8,492/milliliter.

(489) In Example 1, as shown in “Table 1”, at the rate of 8 liters/minute, the diameter of the microbubbles largest in quantity was 29.12 micrometers (μm), the quantity of the microbubbles largest in quantity was 3,918/milliliter, and the total quantity of microbubbles was 4,634/milliliter.

(490) In Example 2, as shown in “Table 1”, the diameter of the microbubbles largest in quantity was 27.92 micrometers (μm), the quantity of the microbubbles largest in quantity was 2,653/milliliter, and the total quantity of microbubbles was 3,509/milliliter.

(491) In Example 3, as shown in “Table 1”, the diameter of the microbubbles largest in quantity was 27.92 micrometers (μm), the quantity of the microbubbles largest in quantity was 4,707/milliliter, and the total quantity of microbubbles was 6,023/milliliter.

(492) In Comparative Example 1, as shown in “Table 1”, the diameter of the microbubbles largest in quantity was 7.19 micrometers (μm), the quantity of the microbubbles largest in quantity was 595/milliliter, and the total quantity of microbubbles was 1,722/milliliter.

(493) In Example 1, Example 2, and Example 3, as compared to Comparative Example 1, the diameter of the microbubbles largest in quantity can be increased.

(494) In Example 1, Example 2, and Example 3, as compared to Comparative Example 1, a sufficient volume of the microbubbles largest in quantity can be mixed into water (liquid). In particular, in Example 1, at the rate of 10 liters/minute, the diameter of the microbubbles largest in quantity was 28.67 micrometers (μm), and the quantity of the microbubbles largest in quantity was 6,060/milliliter. Thus, as compared to Example 2, Example 3, and Comparative Example 1, the sufficient volume of the microbubbles largest in quantity can be mixed into water (liquid), and hence significant effects can be expected.

(495) In Example 1, Example 2, and Example 3, as compared to Comparative Example 1, the sufficient volume of microbubbles can be mixed into water (liquid).

(496) Thus, when the plurality of flow-adjustment-piece plates **116** are formed on the flow-adjustment nozzle disk **114** as in the “flow-adjustment piece” in Example 1, Example 2, and Example 3, the sufficient volume of microbubbles can be mixed into water (liquid).

(497) Measurement results of the ultrafine bubbles in Example 1, Example 2, Example 3, and Comparative Example 1 are shown in “Table 2”.

(498) TABLE-US-00002 TABLE 2 Measurement results of ultrafine bubbles in “shower test”

Diameter of the ultrafine bubbles (nm)	Quantity of the ultrafine bubbles (milliliter)	Total quantity of ultrafine bubbles (milliliter)
Example 1 98	140,000	27,000,000 (10 L/min.)
Example 2 134.5	290,000	5,400,000
Example 3 128.8	160,000	3,800,000
Comparative Example 1 150.8	440,000	6,500,000

Diameter of the ultrafine bubbles: nanometer
Quantity of the ultrafine bubbles largest in quantity: number of the ultrafine bubbles/milliliter
Total quantity of ultrafine bubbles: total number of ultrafine bubbles/milliliter

(499) In Example 1, as shown in “Table 2”, at the rate of 10 liters/minute, the diameter of ultrafine bubbles largest in quantity was 98 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 140,000/milliliter, and the total quantity of ultrafine bubbles was 27,000,000/milliliter.

(500) In Example 1, as shown in “Table 2”, at the rate of 8 liters/minute, the diameter of ultrafine bubbles largest in quantity was 136.9 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 730,000/milliliter, and the total quantity of ultrafine bubbles was 13,000,000/milliliter.

(501) In Example 2, as shown in “Table 2”, the diameter of ultrafine bubbles largest in quantity was 134.5 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 290,000/milliliter, and the total quantity of ultrafine bubbles was 5,400,000/milliliter.

(502) In Example 3, as shown in “Table 2”, the diameter of ultrafine bubbles largest in quantity was 128.8 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 160,000/milliliter, and the total quantity of ultrafine bubbles was 3,800,000/milliliter.

(503) In Comparative Example 1, as shown in “Table 2”, the diameter of ultrafine bubbles largest in quantity was 150.8 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 440,000/milliliter, and the total quantity of ultrafine bubbles was 6,500,000/milliliter.

(504) In Example 1, Example 2, and Example 3, the diameter of ultrafine bubbles largest in quantity was 90 to 136.9 nanometers, and the quantity of ultrafine bubbles largest in quantity was 140,000 to 730,000/milliliter. Thus, the sufficient volume of ultrafine bubbles largest in quantity can be mixed into water (liquid).

(505) In Example 1, Example 2, and Example 3, the total quantity of ultrafine bubbles was 730,000 to 2,700,000/milliliter. Thus, the sufficient volume of ultrafine bubbles can be mixed into water (liquid).

(506) In particular, in Example 1, as compared to Example 2, Example 3, and Comparative Example 1, the sufficient volume of ultrafine bubbles largest in quantity can be mixed into water (liquid).

(507) In Example 1, as compared to Example 2, Example 3, and Comparative Example 1, the sufficient quantity of ultrafine bubbles in total can be mixed into water (liquid).

(508) <2>“Mist Test”

(509) The mist test was carried out in Example 4 and Comparative Example 2.

(510) (1) Mist Throttle Hole

(511) The “mist throttle hole” was common (the same) in Example 4 and Comparative Example 2.

(512) The “mist throttle hole **121** (conical hole)” in Example 4 and Comparative Example 2 is described with reference to FIG. **43** and FIG. **44**.

(513) The “mist throttle hole **121**” in Example 4 has the following configuration. Number of the mist throttle holes **121**: 12 Radius of the circle CK: 18.4 mm Hole diameter dM of each of the mist throttle holes **121**: 0.96 mm (opened in the disk front surface **96A**) Hole diameter dF of each of the mist throttle holes **121**: 4.0 mm (opened in the disk back flat surface **96B**) Hole length of each of the mist throttle holes **121**: 5.8 mm

(514) The mist throttle holes **121** were arranged on the circle CK, and were arranged at equal angular intervals (equal pitches) of 30 degrees in the peripheral direction of the circle CK (or the shower nozzle **3**).

(515) (2) Mist Guide (Conical Spiral) and Guide Ring

(516) The “mist guide **124**” in Example 4 is described with reference to FIG. **52** to FIG. **55**.

(517) The “mist guide **124**” in Example 4 has the following configuration. Number of the mist guides: 12 Number of the spiral surfaces: 2 (first and second spiral surfaces) Guide height GL: 3.5 mm Maximum bottom width GH: 8.95 mm Ring diameter D8 of the circle CL of the guide ring **123**: 18.4 mm

(518) Each of the mist guides **124** was formed integrally with the guide ring **123** so that the cone center line L thereof was located on the circle CL. The mist guides **124** were arranged on the guide ring **123** at equal angular intervals of 30 degrees in the peripheral direction of the circle CL.

(519) Each of the mist guides **124** was inserted into each of the mist throttle holes **121** from the cone upper surface **124A**, and was fitted in each of the mist throttle holes **121** with the gap between the cone side surface **124C** and the conical inner peripheral surface **121A** of each of the mist throttle holes **121**.

(520) Thus, each of the mist guides **124** was fitted to the shower nozzle **3** (or the shower nozzle plate **96**), thereby defining the first and second mist flow passages **51** and **52** between the first and second spiral surfaces **127** and **128** and the conical inner peripheral surface **121A** of each of the mist throttle holes **121**.

(521) In Comparative Example 2, there is employed mist generating means “without a mist guide”, in which the mist guides are not inserted into the mist throttle holes, respectively.

(522) (3) Liquid, Static Liquid Pressure (Hydrostatic Pressure) of Liquid, and Liquid Feeding Rate (Water Feeding Rate)

(523) In Example 4 and Comparative Example 2, the following is employed. Liquid: tap water (water) Static liquid pressure (hydrostatic pressure) of the liquid (water): 0.2 MPa (megapascals) Liquid feeding rate (water feeding rate) of the liquid (water): 7.4 liters/minute (7.4 liters per minute)

(524) In Example 4 and Comparative Example 2, under a condition in which the “hydrostatic pressure” was 0.2 MPa and the “water feeding rate” was 7.4 liters/minute, the tap water was caused to flow into the inflow passage and jetted through the mist throttle holes.

(525) (4) Measurement of Quantity of Air Bubbles

(526) In the “mist test”, a quantity of air bubbles mixed into a mist of water droplets (liquid droplets) jetted through the mist throttle holes was measured.

(527) In Example 4 and Comparative Example 2, a total quantity of the micrometer-sized air bubbles (microbubbles) and a total quantity of the nanometer-sized air bubbles (ultrafine bubbles) were measured in a case in which the mist of water droplets was jetted at a rate of 4 liters/minute.

(528) In Example 4 and Comparative Example 2, the quantity of air bubbles (bubble quantity) contained per a milliliter (ml) of the mist of water droplets was measured.

(529) In Example 4 and Comparative Example 2, a total quantity of ultrafine bubbles and an ultrafine bubble diameter of ultrafine bubbles largest in quantity were measured.

(530) Measurement results of the microbubbles in Example 4 and Comparative Example 2 are shown in “Table 3”.

(531) TABLE-US-00003 TABLE 3 Measurement results of microbubbles in “mist test” Diameter of the Quantity of the microbubbles microbubbles largest in largest in Total quantity quantity quantity of microbubbles Example 4 11.52 21,079 27,022 Comparative 3.24 1,680 2,637 Example 2 Diameter of the microbubbles: micrometer Quantity of the microbubbles largest in quantity: number of the microbubbles/milliliter Total quantity of microbubbles: total number of microbubbles/milliliter

(532) In Example 4, as shown in “Table 3”, the diameter of microbubbles largest in quantity was 11.52 micrometers (μm), the quantity of the microbubbles largest in quantity was 21,079/milliliter, and the total quantity of microbubbles was 27,022/milliliter.

(533) In Comparative Example 2, as shown in “Table 3”, the diameter of the microbubbles largest in quantity was 3.24 micrometers (μm), the quantity of the microbubbles largest in quantity was 1,680/milliliter, and the total quantity of microbubbles was 2,637/milliliter.

(534) In Example 4, as compared to Comparative Example 2, a sufficient volume of the microbubbles largest in quantity can be mixed into the mist of water droplets (the liquid droplets).
(535) In Example 4, as compared to Comparative Example 2, the sufficient quantity of microbubbles in total can be mixed into the mist of water droplets (the liquid droplets).
(536) Thus, in the “mist test”, when the mist guides each having a conical spiral shape (a truncated conical spiral shape) are fitted in the mist throttle holes, respectively, the sufficient volume of microbubbles can be mixed into the mist of water droplets (liquid droplets).

(537) Measurement results of the ultrafine bubbles in Example 4 and Comparative Example 2 are shown in “Table 4”.

(538) TABLE-US-00004 TABLE 4 Measurement results of ultrafine bubbles in “mist test”
Diameter of the ultrafine bubbles largest in quantity: nanometer
Quantity of the ultrafine bubbles largest in quantity: number of the ultrafine bubbles/milliliter
Total quantity of ultrafine bubbles: total number of ultrafine bubbles/milliliter

(539) In Example 4, as shown in “Table 4”, the diameter of ultrafine bubbles largest in quantity was 124.1 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 710,000/milliliter, and the total quantity of ultrafine bubbles was 14,000,000/milliliter.

(540) In Comparative Example 2, as shown in “Table 4”, the diameter of ultrafine bubbles largest in quantity was 128.1 nanometers (nm), the quantity of ultrafine bubbles largest in quantity was 360,000/milliliter, and the total quantity of ultrafine bubbles was 6,600,000/milliliter.

(541) In Example 4, as compared to Comparative Example 2, a sufficient volume of ultrafine bubbles largest in quantity can be mixed into the mist of water droplets (liquid droplets).

(542) In Example 4, as compared to Comparative Example 2, the sufficient quantity of ultrafine bubbles in total can be mixed into the mist of water droplets (liquid droplets).

INDUSTRIAL APPLICABILITY

(543) The present invention is most suitable for jetting the air bubble-liquid mixture or the mist of liquid droplets.

REFERENCE SIGNS LIST

(544) X shower head **1** shower main body **2** flow passage switching means **3** shower nozzle **4** air bubble-liquid mixture generating means **5** mist generating means **9** inflow passage **10** outflow passage **96** shower nozzle plate **97** shower cylindrical portion **98** air bubble-liquid mixture jetting hole **111** flow-adjustment piece **112** air introduction passage **114** flow-adjustment nozzle disk **116** flow-adjustment-piece plate **117** liquid throttle hole BR air bubble mixing space GP mixing gap

Claims

1. A mist generating unit arranged on a shower nozzle plate, and configured to form a liquid having flowed out through an outflow passage into a mist of liquid droplets in which air bubbles are mixed and dissolved, the mist generating unit comprising: a plurality of mist throttle holes, which are formed to pass through the shower nozzle plate having a hole length between a disk front surface and a disk back flat surface of the shower nozzle plate, are opened in the disk front surface and the disk back flat surface of the shower nozzle plate, and communicate from the disk back flat surface of the shower nozzle plate with the outflow passage; a mist ring body; wherein the mist ring body includes a guide ring having a plurality of guide protrusions; wherein the mist ring body includes a plurality of mist guides, which are each formed into a conical spiral shape with a height set smaller than the hole length of each of the mist throttle holes, and each includes first and second spiral surfaces each having the same spiral shape; wherein the guide protrusions are each arranged at intervals in the circumferential direction of the guide ring; wherein the guide protrusions are each formed to protrude in a direction orthogonal to a center line of the guide ring, and are each formed

integrally with the guide ring; wherein the mist throttle holes are each arranged at intervals and each formed into a conical hole which passes through the shower nozzle plate, has a diameter gradually reducing from the disk back flat surface of the shower nozzle plate of the outflow passage side, and is opened in the disk front surface of the shower nozzle plate, wherein the first and second spiral surfaces are arranged between a cone bottom flat surface and a cone upper surface of each of the mist guides to cross a cone side surface of each of the mist guides, wherein the first and second spiral surfaces are arranged so as to be point symmetrical with respect to a cone center line of each of the mist guides, wherein the first spiral surface is formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface along the second spiral surface, wherein the second spiral surface is formed into a spiral shape having a diameter gradually reducing from the cone bottom flat surface toward the cone upper surface along the first spiral surface; wherein each of the mist guides is arranged between each of the guide protrusions at intervals in the peripheral direction of the guide ring; wherein each of the mist guides is arranged so that a surface end of the first spiral surface and a surface end of the second spiral surface are respectively located at an outer peripheral surface and an inner peripheral surface of the guide ring at the cone bottom flat surface; wherein each of the mist guides is fixed integrally with the guide ring so that the cone bottom flat surface is held in abutment against the guide ring; wherein the cone bottom flat surface is fixed to the guide ring so as to protrude from the inner peripheral surface and the outer peripheral surface of the guide ring in the direction orthogonal to the center line of the guide ring; wherein each of the mist guides is inserted into each of the mist throttle holes from the cone upper surface with a gap between the cone side surface and a conical inner peripheral surface of each of the mist throttle holes, wherein each of the mist guides defines a first mist flow passage having a spiral shape between the first spiral surface, the conical inner peripheral surface, and the cone side surface, wherein each of the mist guides defines a second mist flow passage having a spiral shape between the second spiral surface, the conical inner peripheral surface, and the cone side surface; wherein along with insertion of the mist guides into the mist throttle holes, the guide protrusions are each brought into abutment against the disk back flat surface of the shower nozzle plate; wherein the first mist flow passage is defined in a spiral shape to extend along the second mist flow passage from the cone bottom flat surface toward the cone upper surface, communicates with the outflow passage, and is opened into each of the mist throttle holes between the disk front surface of the shower nozzle plate and the cone upper surface, wherein the second mist flow passage is defined in a spiral shape to extend along the first mist flow passage from the cone bottom flat surface toward the cone upper surface, communicates with the outflow passage, and is opened into each of the mist throttle holes between the disk front surface of the shower nozzle plate and the cone upper surface, wherein the first and second mist flow passages jet the liquid flowing through each of the mist flow passages into each of the mist throttle holes between the disk front surface of the shower nozzle plate and the cone upper surface to jet a mist of liquid droplets in which ultrafine bubbles are mixed and dissolved from each of the mist throttle holes.
